Policy Rule Regressions with Survey Data[†]

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Abstract

Endogeneity bias is ubiquitous and can be sizable in the ordinary least squares (OLS) estimation of policy rule coefficients. This paper introduces a simple procedure that leverages survey data to rectify the bias under flexible information assumptions. We decompose policy rule regressors (e.g., inflation and output gap) into their forecasts made before policy decisions and the associated forecast errors. The forecasts are readily available in survey data and, by construction, orthogonal to the forecast errors and policy shocks under complete information. We further orthogonalize the forecast errors to remove the bias in the presence of information rigidities. Using Monte Carlo simulations, we showcase the efficacy of this procedure in a prototypical new Keynesian model under distinct information structures and policy regimes. As an empirical application, we employ Bayesian methods to compare the performance of the standard OLS approach using real-time data and our approach based on survey data in estimating monetary and fiscal policy rules. Marginal likelihood estimates reveal that our approach consistently outperforms across nearly all model specifications considered.

Keywords: Endogeneity bias; Incomplete information; Two-stage OLS; Survey data.

JEL Classification: C11, C22, E52

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1 Introduction

The policy reaction functions of central banks and governments are often summarized by straightforward rules. A prominent illustration is the Taylor rule, delineated in Taylor (1993), which posits that the central bank sets short-term interest rates contingent upon variations in the inflation rate and the output gap. However, estimating such reaction functions using ordinary least squares (OLS) often introduces an endogeneity bias. This bias occurs because central banks adjust their policies in response to economic variables that are simultaneously influenced by policy shocks.

Theoretically, the endogeneity bias in estimating policy reaction functions can be addressed using instrumental variables or the generalized method of moments (GMM). In practice, however, identifying valid and strong instruments can be challenging. A common approach in the literature involves using lagged endogenous variables as instruments. Yet, this strategy may be problematic when policy rules exhibit inertia. This is because the same lagged variables may appear repeatedly in the specification, thereby introducing multicollinearity concerns. This issue is especially severe in "passive" monetary policy regimes, in which the central bank responds only weakly to inflation. A large literature documents such regimes in the U.S. prior to the Volcker disinflation [the pre-1979 period; see, for example, Clarida et al. (2000)]. In these environments, interest rate inertia dominates the dynamics of the policy rate, so lagged interest rates absorb much of the variation that would otherwise be attributed to lagged inflation and the output gap. Consequently, the lags of inflation and output gap that enter both the policy rule and the instrument set become highly collinear.

Under a complete-information framework, Carvalho et al. (2021) argue in favor of OLS estimation of policy rules despite this endogeneity bias. Nevertheless, a growing body of work examines the implications of departures from the full-information setting. Coibion and Gorodnichenko (2015) and Angeletos et al. (2021), for instance, provide empirical evidence supporting the presence of incomplete information. These information frictions can introduce additional biases into the estimated coefficients, raising the natural question of how to mitigate such biases under flexible information settings.

We introduce a simple method using survey data that effectively reduces the endogeneity bias, inspired by the identification strategy proposed by Romer and Romer (2004).[1] Under complete information, we decompose policy rule regressors (e.g., inflation and output gap) into two components:

[1] Romer and Romer (2004) regress intended funds-rate changes on the Federal Reserve's internal Greenbook forecasts and interpret the residual as an exogenous monetary policy shock. By conditioning on staff or survey forecasts of inflation and the output gap, this procedure strips out the systematic component of policy that reflects the central bank's response to expected macroeconomic conditions, thereby mitigating the endogeneity between the policy instrument and realized outcomes. Related work also uses survey-based expectations directly in Taylor-rule-type regressions to address endogeneity in expectations equations [see, for example, Coibion and Gorodnichenko (2011) and Coibion et al. (2018)]. In contrast, our methods use survey data as a proxy to reduce the endogeneity bias when estimating contemporaneous data-based Taylor rules, without requiring expectations to enter explicitly as regressors, and they perform well in flexible-information environments.

forecasts derived from the previous period and their associated forecast errors. The forecasts, by construction, are orthogonal to both the forecast errors and policy shocks under complete information. This orthogonality arises because forecast errors reflect current period information, whereas forecasts are based on previous period information. We refer to this approach as the one-step method.

Although survey data forecasters are among the most informed economic agents, information rigidities may exist. If agents form expectations under incomplete information, the average forecast errors across agents are not orthogonal to the average forecasts of policy rule regressors, which reintroduces endogeneity bias when employing our one-step method. In this case, we further orthogonalize the forecast errors using the empirical specification suggested by Coibion and Gorodnichenko (2015) (CG henceforth). We refer to this approach as the two-step method. In the first step, we employ a regression of forecast errors on ex-ante mean forecast revisions to obtain their predicted values. In the second step, we create new regressors by summing these predicted values and the forecasts, ensuring the new regressors are orthogonal to the error term.

Our approach offers several advantages. First, it addresses the endogeneity bias in estimating policy rules while yielding robust estimates regardless of whether policymakers rely on contemporaneous data or forecasts in decision-making [see also Orphanides (2003) and Coibion and Gorodnichenko (2012)]. Second, forecasts for key macroeconomic variables are readily available from the U.S. Survey of Professional Forecasters (SPF) and the Tealbook Data Set. Third, our method leverages earlier period information, thereby avoiding the common approach of using lagged variables as instrumental variables.

To assess the efficacy of our method, we first quantify the estimation bias in dynamic stochastic general equilibrium (DSGE) models and investigate how expectations data can help reduce this bias under various information structures and policy regimes. Using a standard new Keynesian model under complete information, our simulation results reveal that the conventional OLS approach can yield highly inaccurate estimates and might even misidentify policy regimes, while the use of expectations data substantially improves these estimates. We also examine a sticky information model based on Reis (2009), where only a fraction of agents update their information each period. The simulation results demonstrate that the OLS approach using real-time data leads to biased estimates, whereas our methods significantly improve the estimates, with the two-step method outperforming the one-step approach.

We then conduct empirical analysis using a Bayesian framework to compare the performance of standard OLS with our proposed methods for estimating monetary and fiscal policy rules. Using inflation and output gap forecasts from the SPF and Tealbook dataset, our sample covers the period from the fourth quarter of 1987 to the fourth quarter of 2006, just before the Great Recession in

2007.[2] During this period, we cannot reject the null hypothesis of complete information, implying that information rigidities do not pose a serious issue. Therefore, the one-step method is sufficient for our purposes. The Bayesian results show that our approaches consistently yield higher marginal likelihoods compared to standard OLS estimation across nearly all model specifications.

Our paper makes three main contributions. First, we propose simple methods that leverage survey data to rectify the bias under flexible information assumptions. Importantly, we point out that the use of survey forecasts may still be problematic in the presence of information frictions. We then propose the two-step method to "adjust" regressors, leveraging the pattern of forecast revisions documented by Coibion and Gorodnichenko (2015). Second, we adapt the Bayesian model scan framework of Chib and Zeng (2020) to compare log marginal likelihoods across competing specifications, providing a coherent, fully Bayesian ranking of non-nested models and selecting the specification that is asymptotically closest to the true data-generating process. Third, we show that our methods yield reliable GMM instruments under flexible information settings and policy regimes: relative to conventional lagged endogenous instruments, survey-based lagged-information instruments consistently reduce the endogeneity bias. Our procedures can be interpreted as method-of-moments estimators.

The rest of the paper is organized as follows. In Section 2, we introduce our method to address the endogeneity issue in policy rule regressions. Section 3 quantifies how the proposed method reduces the endogeneity bias under different information settings. Section 4 presents the empirical application of our method. Section 5 concludes the paper.

2 Policy Rule Regressions

In this section, we propose empirical methods that use survey data in estimating policy rules to eliminate the endogeneity bias under various information settings. Consider the following general autoregressive processes

$$y_t = \sum_{k=1}^{K} \rho_k y_{t-k} + \sum_{i=1}^{I} \beta_i X_{i,t} + \varepsilon_t, \tag{2.1}$$

where y_t represents the policy choice, ρ_k measures the policy smoothing on the k-th lagged term, β_i represents the coefficient on explanatory variable $X_{i,t}$ (e.g., inflation and output gap in a Taylor rule), and the exogenous shock ε_t is assumed to be independently and identically distributed. An endogeneity issue may arise when estimating β_i using OLS regression, as the policy rule regressor $X_{i,t}$ could be correlated with the error term ε_t. To address this endogeneity issue, we decompose the policy rule regressors to separate the new explanatory variables from the new error term based on the timing of information arrival. This decomposition ensures that the two components remain orthogonal.

[2]The Great Recession is widely considered as a structural break in the literature [see, e.g., Bunzel and Enders (2010)].

Specifically, we decompose $X_{i,t}$ into two components, i.e., forecasts and the associated forecast errors, and rewrite (2.1) as

$$y_t = \sum_{k=1}^{K} \rho_k y_{t-k} + \sum_{i=1}^{I} \beta_i \mathbb{E}_{t-1} X_{i,t} + \underbrace{\sum_{i=1}^{I} \beta_i [X_{i,t} - \mathbb{E}_{t-1} X_{i,t}] + \varepsilon_t}_{\text{new error term}}, \tag{2.2}$$

where $\mathbb{E}_{t-1} X_{i,t}$ represents the ex-ante forecast of $X_{i,t}$ and $X_{i,t} - \mathbb{E}_{t-1} X_{i,t}$ captures the corresponding ex-post forecast error.[3] It is important to note that ex-ante forecasts are formed using information available at period $t-1$ and earlier. In other words, $\mathbb{E}_{t-1} X_{i,t}$ is independent of any information from period t. Meanwhile, the forecast error is linked to information available at time t. Similarly, ε_t also depends on time t information. Therefore, the new error term, $\sum_{i=1}^{I} \beta_i [X_{i,t} - \mathbb{E}_{t-1} X_{i,t}] + \varepsilon_t$, is determined exclusively by information available at time t. This decomposition effectively helps mitigate the endogeneity issue in parameter estimation.

We consider the Taylor rule as an example. In a Taylor-type interest rate reaction function, monetary policymakers respond to real-time inflation and output gap as follows

$$r_t = \alpha + \psi_1 \pi_t + \psi_2 ygap_t + \varepsilon_t, \tag{2.3}$$

where r_t denotes the nominal interest rate, π_t represents the inflation rate, $ygap_t$ is the output gap at time t, and ψ_1 and ψ_2 are the policy coefficients that need to be estimated. Given that both π_t and $ygap_t$ may be correlated with ε_t, we apply the decomposition in (2.2) and rewrite (2.3) as

$$r_t = \alpha + \psi_1 \mathbb{E}_{t-1} \pi_t + \psi_2 \mathbb{E}_{t-1} ygap_t + \underbrace{\psi_1 [\pi_t - \mathbb{E}_{t-1} \pi_t] + \psi_2 [ygap_t - \mathbb{E}_{t-1} ygap_t] + \varepsilon_t}_{\text{new error term}}, \tag{2.4}$$

where $\mathbb{E}_{t-1} \pi_t$ and $\mathbb{E}_{t-1} ygap_t$ represent the inflation and output gap ex-ante forecasts based on information available at $t-1$, respectively.

Given this specification, we can use survey data for parameter estimation, as it provides direct measures of forecasts.[4] For example, inflation forecasts are available from the Survey of Professional Forecasters, a survey of approximately 40 experts conducted by the Federal Reserve Bank of Philadelphia. Moreover, output gap forecasts can be obtained from the Tealbook (formerly the Greenbook) dataset, which contains real-time estimates and projections of the output gap used by the Board of Governors of the Federal Reserve System. We refer to this approach as the one-step method.

However, the orthogonality between the regressors and the error term in (2.2) may break down

[3] Equation (2.2) is algebraic decompositions of Equation (2.1).

[4] Our specification (2.4) is fundamentally different from Coibion and Gorodnichenko (2012), which assumes that policymakers target forecasts instead of real-time variables.

in the presence of information rigidity. As CG demonstrate, ex-post forecast errors can be correlated with ex-ante forecast revisions.[5] In essence, average forecast errors across agents may reflect information from the previous period, creating a correlation between the error term and ex-ante forecasts. As a result, applying the OLS method to (2.2) may fail to eliminate the endogeneity bias.

To address this issue, we further orthogonalize average forecast errors on average forecast revisions. Specifically, we estimate

$$\underbrace{X_{i,t} - \mathbb{E}_{t-1}X_{i,t}}_{\Delta X_{i,t}} = \alpha_i + \beta_i(\mathbb{E}_{t-1}X_{i,t} - \mathbb{E}_{t-2}X_{i,t}) + \varepsilon_{i,t}. \tag{2.5}$$

We then obtain the predicted value of $\Delta X_{i,t}$, denoted as $\Delta\hat{X}_{i,t}$, and rewrite (2.2) as

$$y_t = \sum_{k=1}^{K} \rho_k y_{t-k} + \sum_{i=1}^{I} \beta_i \underbrace{[\mathbb{E}_{t-1}X_{i,t} + \Delta\hat{X}_{i,t}]}_{\text{new policy rule regressor}} + \varepsilon_t, \tag{2.6}$$

where the new policy rule regressor, $\mathbb{E}_{t-1}X_{i,t} + \Delta\hat{X}_{i,t}$, is by construction based on information available at time $t-1$, while the error term ε_t remains independent of information at $t-1$. The endogeneity bias is thus removed, allowing us to use the OLS method to estimate the coefficients in (2.6). We refer to this approach as the two-step method. In the first step, we regress ex-post forecast errors on ex-ante forecast revisions to obtain their predicted values. In the second step, we construct new policy rule regressors by summing these predicted forecast errors with the ex-ante forecasts. Finally, we estimate the policy rule using these newly constructed regressors.

3 Simulations

In this section, we quantify the endogeneity bias in single-equation estimations of policy rules within DSGE models under various information structures and policy regimes. Our simulation results show that the one-step method significantly reduces the endogeneity bias under complete information, while the two-step method further mitigates the bias in the presence of information rigidity.

3.1 Complete-Information Model

We first demonstrate how using survey data can reduce the endogeneity bias in a complete-information setting. We consider a textbook version of the new Keynesian model presented in Woodford (2003a) and Galí (2008), augmented with a fiscal policy rule. The model consists of a representative household and a continuum of firms, each producing a differentiated product; only a fraction of firms can adjust their prices each period; a cashless economy with one-period nominal bonds B_t selling at a price of $1/R_t$, where R_t is the monetary

[5] CG propose a novel approach to test the complete information hypothesis. They introduce two models of information rigidity: the sticky-information model and the noisy-information model. Both models exhibit a similar empirical relationship between ex-post mean forecast errors and ex-ante mean forecast revisions.

policy instrument; a primary surplus s_t with lump-sum taxation and zero government spending, resulting in consumption being equal to output, $c_t = y_t$; a monetary authority and fiscal authority.

Let $\hat{x}_t \equiv \ln(x_t) - \ln(x)$ represent the log-deviation of a generic variable x_t from its steady state x. A log-linear approximation of the model's equilibrium conditions around the zero-inflation steady state yields

$$\text{Dynamic IS equation:} \quad \hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \tau^{-1}(\hat{R}_t - \mathbb{E}_t\hat{\pi}_{t+1} - \mathbb{E}_t\hat{z}_{t+1}) \tag{3.1}$$

$$\text{New Keynesian Phillips curve:} \quad \hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa\hat{y}_t + \kappa\tau^{-1}\hat{u}_t \tag{3.2}$$

$$\text{Monetary policy:} \quad \hat{R}_t = \rho_R\hat{R}_{t-1} + (1-\rho_R)(\psi_\pi\hat{\pi}_t + \psi_y\hat{y}_t) + \epsilon_{R,t} \tag{3.3}$$

$$\text{Fiscal policy:} \quad \hat{s}_t = \rho_s\hat{s}_{t-1} + (1-\rho_s)\psi_s\hat{b}_{t-1} + \epsilon_{s,t} \tag{3.4}$$

$$\text{Government budget constraint:} \quad \hat{b}_t = \beta^{-1}(\hat{b}_{t-1} + \hat{R}_{t-1} - \hat{\pi}_t - \hat{z}_t) - (\beta^{-1} - 1)\hat{s}_t \tag{3.5}$$

$$\text{Technology shock:} \quad \hat{z}_t = \rho_z\hat{z}_{t-1} + \epsilon_{z,t} \tag{3.6}$$

$$\text{Markup shock:} \quad \hat{u}_t = \rho_u\hat{u}_{t-1} + \epsilon_{u,t} \tag{3.7}$$

where $\pi_t = P_t/P_{t-1}$ is the inflation between periods $t-1$ and t, and $b_t = B_t/P_t$ is the real debt at the end of period t. Equations (3.1)–(3.3) form the key building blocks of the standard new Keynesian model, (3.4) is the model analog to many surplus-debt regression studies that aim to test for fiscal sustainability, and (3.5) is the log-linearized version of the government's flow budget identity, $\frac{1}{R_t}\frac{B_t}{P_t} + s_t = \frac{B_{t-1}}{P_t}$. Together, (3.1)–(3.5) form a linear expectational difference equation system, capturing the core dynamics of most monetary DSGE models.

As is well-known in the literature, this model has two determinate monetary-fiscal policy regimes, and empirical identification of these regimes has received considerable attention [see Leeper and Leith (2016) for an overview]. Misestimating monetary and fiscal policy coefficients can distort the transmission channels and the effects of policy shocks, potentially leading to incorrect impulse responses in DSGE model simulations. In the fiscal dominance regime, we set $\psi_\pi < 1$ and $\psi_s = 0$—the fiscal authority refrains from adjusting its primary balance to stabilize public debt, thereby requiring the monetary authority to accommodate fiscal imbalances. In the monetary dominance regime, we set $\psi_\pi > 1$ and $\psi_s = 0.5$—the monetary authority responds actively to inflation, while the fiscal authority adjusts its primary balance to passively ensure debt sustainability. Other model parameters are calibrated to standard values from the literature, as shown in Table 1.

To assess the effectiveness of our method compared with traditional OLS and GMM approaches, we conduct 6,000 Monte Carlo simulations and discard the first 1,000 draws as burn-in. For the OLS approach, we regress the policy rule on the lagged policy rate and contemporaneous inflation and the output gap. For GMM estimation, we follow standard practice in the literature by using lagged endogenous variables as instruments—specifically, the one-period and two-period lags of inflation and the output gap. However, GMM estimation faces several practical challenges. First, identifying valid and strong instruments is inherently difficult. While lagged endogenous variables

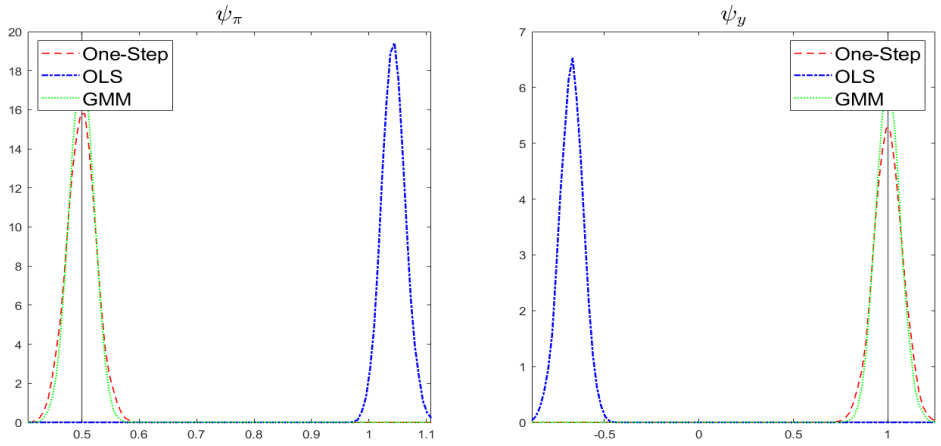
Table 1: Parameters

Parameter	Description	value
τ	elasticity of intertemporal substitution	2.00
κ	the slope of the new Keynesian Philips curve	0.20
β	discount factor	0.99
ψ_y	coefficients of Taylor rule on output gap	1.00
ρ_R	monetary policy smoothing	0.50
ρ_s	fiscal policy smoothing	0.50
ρ_z	AR(1) coefficient on technology shock	0.60
ρ_u	AR(1) coefficient on Markup shock	0.70
σ_R	standard deviations on monetary policy shock	0.15
σ_s	standard deviations on fiscal policy shock	1.50
σ_z	standard deviations on technology shock	0.40
σ_u	standard deviations on markup shock	0.50

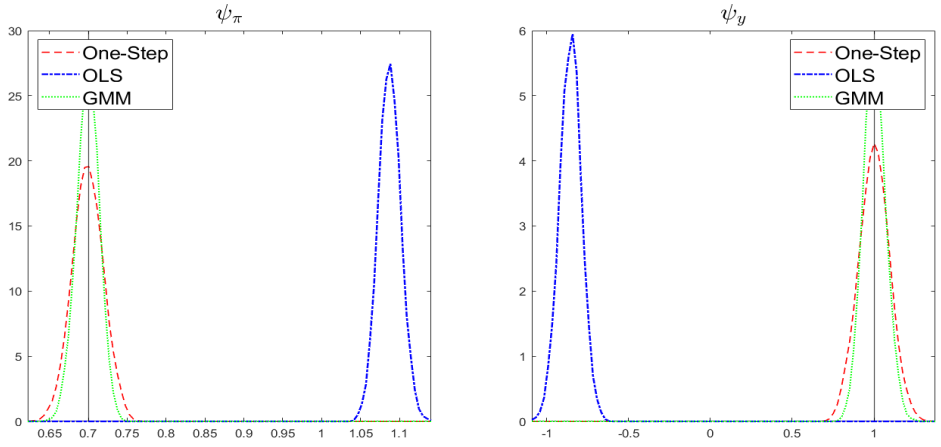
are commonly employed, they can be problematic when policy rules exhibit inertia, as these variables may appear multiple times in the specification, potentially inducing multicollinearity. This issue is especially severe in "passive" monetary policy regimes, in which the central bank responds only weakly to inflation. In such settings, interest rate inertia dominates the dynamics of the policy rate, so lagged interest rates absorb much of the variation that would otherwise load on lagged inflation and the output gap, rendering the lags of inflation and the output gap in the policy rule and the instrument set highly collinear.

Importantly, our proposed methods provide reliable instruments under flexible information settings and policy regimes: relative to conventional lagged endogenous instruments, survey-based lagged-information instruments consistently reduce the endogeneity bias. Our procedures can be interpreted as method-of-moments estimators. We do not claim that our approaches uniformly outperform GMM; rather, we provide GMM with stronger instruments. In simulations, GMM estimates using our instruments with one- and two-period information lags in the output gap and inflation nearly coincide with those from our proposed method. This suggests that a one-period information lag is sufficient to obtain reliable mean estimates.

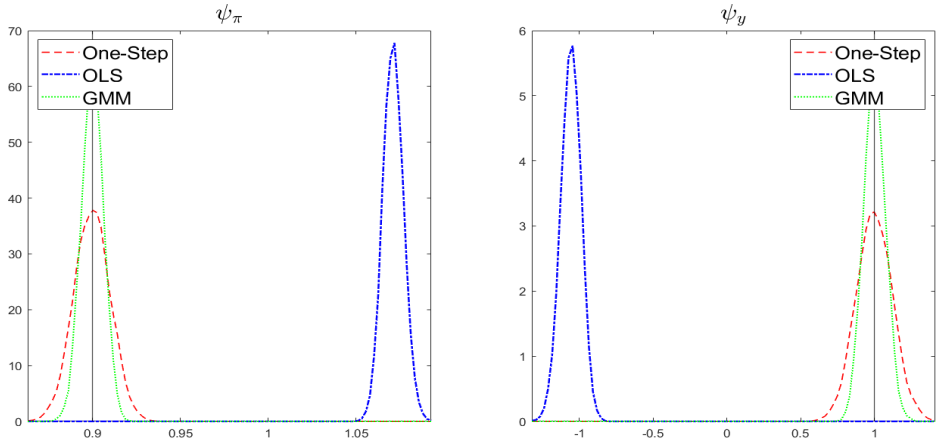
3.1.1 Fiscal Dominance Regime Figure 1 reports the simulation results under the fiscal dominance regime. The blue dashed-dotted line represents the distribution estimated by the OLS method, the green dotted line plots the distribution obtained from the GMM method, and the red dashed line corresponds to the distribution generated by our one-step method. Each simulation was repeated 5000 times and the vertical black solid lines indicate the true parameter values.



(a) $\psi_\pi = 0.5$



(b) $\psi_\pi = 0.7$



(c) $\psi_\pi = 0.9$

Figure 1: Monte Carlo distributions of parameter estimates (OLS, GMM, one-step) under full information in the fiscal-dominance regime

Notes: The blue dashed-dotted line represents the distribution estimated by the OLS method, the green dotted line plots the distribution obtained from the GMM method, and the red dashed line corresponds to the distribution generated by the one-step method. For GMM, we use one-period and two-period lags of inflation and the output gap as instruments. The remaining policy parameters are fixed at $\psi_y = 1$ and $\psi_s = 0$. Distributions are based on 5,000 Monte Carlo replications: we run 6,000 simulations and discard the first 1,000 as burn-in. The results indicate that mean OLS estimates can be substantially biased, whereas our one-step method performs similarly to GMM and is close to the true parameters.

We observe that the OLS estimates exhibit significant bias, with their means deviating substantially from the true parameter values. In contrast, the one-step method based on expectations data and the GMM method yield estimates that are closely aligned with the true coefficients.[6] It is not surprising that GMM performs well in this setting, as lagged inflation and the output gap serve as valid instruments under the assumed data-generating process. However, these instruments can become weak under alternative model configurations. For instance, GMM estimates can be noticeably biased when fiscal shocks have highly persistent effects, causing lagged output gap and inflation to contain information about the fiscal shock. Consequently, these lags become correlated with the structural error term in the policy reaction equation, highlighting the regime-dependent reliability of these instruments.

The OLS estimates also present two significant concerns. First, they often exhibit substantial bias, rendering the estimated policy rules unreliable. Second, this bias may lead to misidentification of the underlying policy regime, posing serious complications for the analysis of monetary-fiscal policy interactions. Such misidentification generates impulse response functions that deviate substantially from the true underlying dynamics, distorting the interpretation of transmission mechanisms and undermining the credibility of policy recommendations. In our fiscal dominance experiment—where $\psi_\pi < 1$ and $\psi_s = 0$—the mean OLS estimates of ψ_π consistently exceed one, erroneously suggesting an active monetary policy stance. This is problematic because specifying both fiscal and monetary policies as "active" leads to no solution in a standard new Keynesian framework.[7] Our one-step method effectively avoids this issue by consistently producing estimates close to the true values across all scenarios.

3.1.2 Monetary Dominance Regime

Figure 2 reports the simulation results under the monetary dominance regime.[8] The one-step and GMM consistently yield more accurate mean estimates of the monetary policy coefficients compared with OLS. Moreover, the distribution of estimates from the one-step method exhibits a lower variance than that of GMM, indicating greater efficiency. Consistent with the findings under the fiscal dominance regime, OLS estimation can also lead to misidentification in the monetary dominance regime. Specifically, OLS systematically underestimates ψ_π in all cases where the true values of ψ_π are greater than unity, leading to consistent

[6] The variance of the one-step method estimator is greater, which aligns with intuition, as survey data inherently exhibit greater uncertainty compared to real-time data. Nevertheless, the findings show that, despite significant heterogeneity in individual information reception, estimation using survey data remains highly effective on average.

[7] Carvalho et al. (2021) reports the impulse response functions (IRFs) of the output gap and inflation to a monetary policy shock. In their experiments, IRFs obtained under Taylor rules estimated by both OLS and GMM are close to the true IRFs. In our settings, however, the OLS estimates are far from the true parameters and there are no IRFs with the mean OLS estimates. OLS estimates are not only far from true values, but also place the economy in a region with no equilibrium solution, so IRFs are not well-defined. We do not claim that our approaches uniformly outperform GMM; rather, we provide GMM with stronger instruments. For this reason, we do not include a direct IRF comparison between GMM and our method in this section.

[8] To maintain consistency with the previous case and ensure the same distance from the boundary value of 1, the coefficient of ψ_π is set to 1.1, 1.3 and 1.5 under the monetary dominance regime.

misclassification of active monetary policy as passive. This misidentification is problematic because when both fiscal and monetary policies are passive, the model becomes indeterminate—there is no unique rational expectations equilibrium.

These results indicate that, although monetary policy shocks are relatively small, the resulting bias can be amplified by fiscal policy shocks. In a simple three-equation New Keynesian model, the asymptotic OLS bias is proportional to the fraction of the variance of the Taylor-rule regressors explained by monetary policy shocks, as emphasized by Carvalho et al. (2021).[9] In a richer environment with both monetary and fiscal policy, interactions with fiscal shocks (and other disturbances) can magnify the influence of even small monetary shocks, leading to sizable estimation bias.

Our results show that the one-step method outperforms OLS in two aspects. First, it consistently produces estimates close to the true values, even in a more complicated framework with both monetary and fiscal policy. Second, it avoids regime misidentification. These findings demonstrate the effectiveness of the decomposition in (2.4), indicating that using survey data increases the accuracy of policy coefficient estimates. In environments with multiple shocks, our methods therefore provide consistent and reliable estimates—even if they exhibit relatively larger variance—while avoiding large biases and incorrect regime classification.

3.2 Sticky-Information Model

Due to information rigidity, ex-post forecast errors may be correlated with ex-ante forecasts, leading to the potential endogeneity bias in the one-step method using survey data. To address this issue, we adopt the two-step method, as outlined in (2.5), to construct new policy rule regressors and further mitigate the bias.

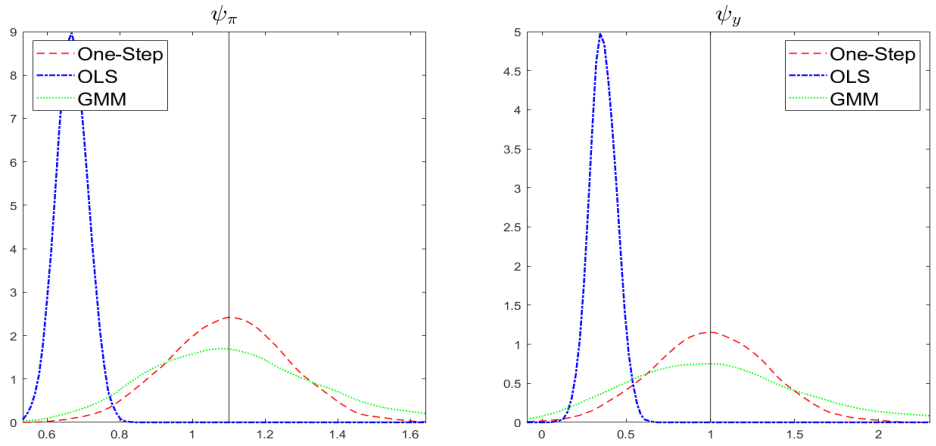
We consider the sticky-information model as presented in Reis (2009).[10] In the goods markets, inattentive firms update their information at rate λ, while $1-\lambda$ fraction of firms keep one-period old information. The core distinction between the complete information and the sticky information models stems from the following Phillips curve

$$p_t = \lambda \sum_{i=0}^{\infty} (1-\lambda)^i \mathbb{E}_{t-i} \left\{ \frac{\beta(w_t - p_t) + (1-\beta) y_t - a_t}{\beta + \nu(1-\beta)} - \frac{\beta}{(\nu-1)[\beta + \nu(1-\beta)]} \nu_t \right\}, \tag{3.8}$$

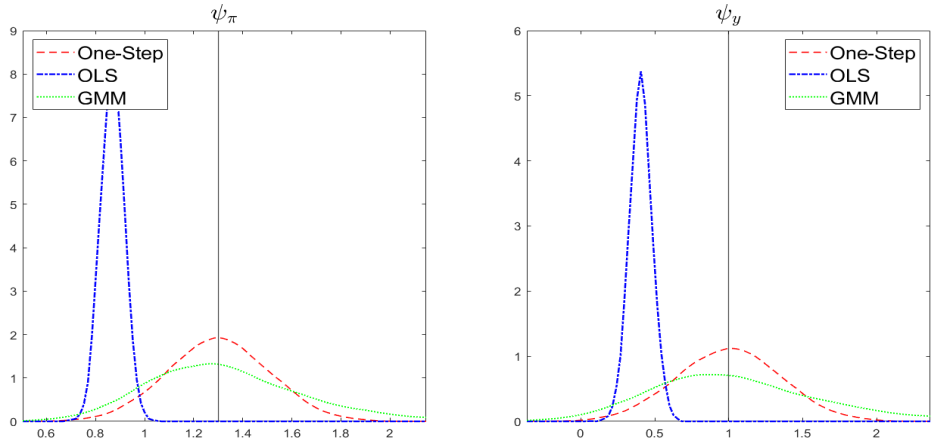
where p_t is the price level, $w_t - p_t$ is the real wage, y_t is the output, a_t represents the productivity shock, and ν_t denotes the random substitutability of varieties of goods with ν being the steady-state elasticity of substitution for goods. The parameter β denotes the labor share. Other structural equations in Reis (2009) include an IS curve, a wage-setting equation, a conventional production

[9] Though Carvalho et al. (2021) consider only monetary policy in both their baseline and extended models, we are still able to replicate their two core findings. First, the variance of monetary policy shocks affects the estimation performance of both OLS and GMM. Second, as this variance increases, GMM tends to outperform OLS (see Appendix C). This confirms that the substantial bias in our OLS estimates does not arise from unusually large monetary policy shocks.

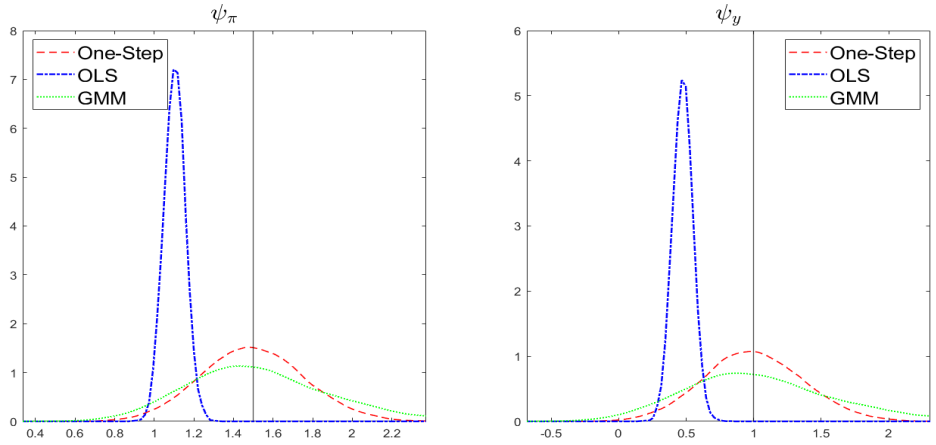
[10] We use sticky-information to characterize information rigidity because the endogeneity problem in parameter estimation stems from the interaction of information across different time periods.



(a) $\psi_\pi = 1.05$



(b) $\psi_\pi = 1.2$



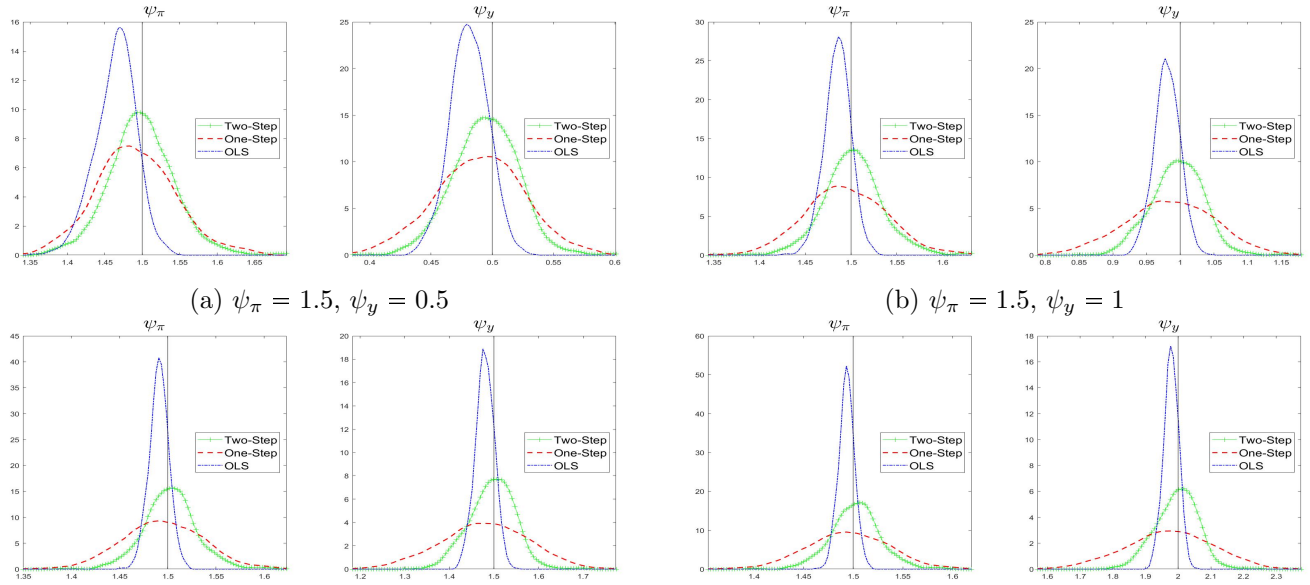
(c) $\psi_\pi = 1.5$

Figure 2: Monte Carlo distributions of parameter estimates (OLS, GMM, one-step) under full information in the monetary-dominance regime

Notes: The blue dashed-dotted line represents the distribution estimated by the OLS method, the green dotted line plots the distribution obtained from GMM, and the red dashed line corresponds to the distribution generated by the one-step method. For GMM, we use one- and two-period lags of inflation and the output gap as instruments. The remaining policy parameters are fixed at $\psi_y = 1$ and $\psi_s = 0.5$. Distributions are based on 5,000 Monte Carlo replications: we run 6,000 simulations and discard the first 1,000 as burn-in. The results indicate that mean OLS estimates can be substantially biased, whereas our one-step method performs similarly to GMM and is close to the true parameters.

function, and a Taylor rule for monetary policy.

We adopt the solution method proposed by Verona and Wolters (2014) and conduct 500 simulations using their calibrated parameters. Figure 3 compares the performance of OLS estimation, the one-step method, and the two-step method. The dashed-dotted blue line represents the distribution from the OLS method, the dashed red line corresponds to the one-step method, and the green line with plus signs illustrates the distribution from the two-step method. The results show that the performance of the one-step method deteriorates significantly in the presence of information rigidity. This occurs because ex-post forecast errors are correlated with ex-ante forecasts, undermining the effectiveness of directly using survey data for parameter estimation. In contrast, the two-step method, which projects average forecast errors onto average forecast revisions as outlined in (2.5), effectively mitigates the endogeneity bias introduced by sticky information. As a result, the mean of the two-step estimator distribution is consistently closer to the true value across all regimes.[11]



(a) $\psi_\pi = 1.5$, $\psi_y = 0.5$ (b) $\psi_\pi = 1.5$, $\psi_y = 1$

(c) $\psi_\pi = 1.5$, $\psi_y = 1.5$ (d) $\psi_\pi = 1.5$, $\psi_y = 2$

Figure 3: Monte Carlo distributions of parameter estimates (OLS, one-step, two-step) under sticky information setting

Notes: This figure illustrates the distribution of estimated policy coefficients (ψ_π and ψ_y) obtained through various methods. The dashed-dotted blue line represents the distribution from the OLS method, the dashed red line corresponds to the one-step method, and the green line marked with plus signs illustrates the distribution from the two-step method. Distributions are based on 500 Monte Carlo simulations. The results show that the two-step method, which projects average forecast errors onto average forecast revisions as outlined in (2.5), effectively mitigates the endogeneity bias introduced by sticky information.

[11] Consistent with the complete-information case, the variance of estimates using survey data remains higher than that using real-time data, reflecting the greater uncertainty inherent in expectational variables.

4 Empirical Analysis

In this section, we apply our proposed methods using survey data to estimate monetary and fiscal policy rules and compare their performance with traditional OLS estimates. In our main empirical specifications, forecasts of inflation and the output gap are taken from the Survey of Professional Forecasters (SPF) and the Tealbook (formerly the Greenbook), respectively. Real-time inflation is computed from the implicit GDP deflator produced by the Bureau of Economic Analysis (BEA) via Archival Federal Reserve Economic Data (ALFRED).[12] Government spending and debt are also produced by the BEA via ALFRED. For the monetary-policy rule, the baseline sample is 1987:Q4–2006:Q4, starting from the earliest data available and ending prior to the Great Recession, which is commonly viewed as a structural break. Robustness checks include: (i) a post-ZLB, pre-pandemic window (2016-2019); (ii) replication with core PCE and core CPI where SPF forecasts are available; and (iii) alternative output-gap constructions using the Real-Time Data Set for Macroeconomists (RTDSM) GDP with the Congressional Budget Office (CBO) potential. We also estimate a fiscal-policy (government spending) rule over the sample period 1987:Q4–2019:Q4.[13] Data details are provided in Appendix A.

4.1 Bayesian Model Scan Approach We adopt the Bayesian model scan of Chib and Zeng (2020) to compare alternative restrictions of the specification (2.1), because the resulting log marginal likelihoods provide a coherent, fully Bayesian ranking of a set of non-nested models, which selects the specification that is asymptotically closest to the true data-generating process. Specifically, the parameters of the general empirical specification (2.1) are given as

$$\boldsymbol{\theta} = (\rho_1, \ldots, \rho_K, \beta_1, \ldots, \beta_I, \sigma). \tag{4.1}$$

Assume that we have J models denoted by $\mathcal{M}_1, \ldots, \mathcal{M}_J$. Let $\Pr(\mathcal{M}_j)$ represent the prior probability of model $\mathcal{M}_j$ on the model space and $m(\mathbf{y}_{1:T}|\mathcal{M}_j, \mathbf{X}_{1:T})$ denote the marginal likelihood of model $\mathcal{M}_j$, where $\mathbf{y}_{1:T}$ is the sample data for the dependent variable and $\mathbf{X}_{1:T}$ corresponds to the sample data for explanatory variables over T time periods. According to the Bayes theorem, the posterior probability of $\mathcal{M}_j$ is given by

$$\Pr(\mathcal{M}_j|y_{1:T}, X_{1:T}) = \frac{\Pr(\mathcal{M}_j)m(\mathbf{y}_{1:T}|\mathcal{M}_j, \mathbf{X}_{1:T})}{\sum_{l=1}^{J} \Pr(\mathcal{M}_l)m(\mathbf{y}_{1:T}|\mathcal{M}_l, \mathbf{X}_{1:T})}, \tag{4.2}$$

[12] We use the GDP implicit price deflator rather than core PCE or core CPI in the main analysis because SPF forecasts for core measures begin only in 2007, whereas our baseline sample spans 1987:Q4–2006:Q4.

[13] The sample period is selected based on the availability of output-gap forecasts in the Tealbook (formerly the Greenbook).

where $m(\mathbf{y}_{1:T}|\mathcal{M}_j, \mathbf{X}_{1:T})$ is calculated numerically using the method proposed by Chib (1995). In Bayesian analysis, models with higher marginal likelihoods are preferred.[14]

We consider the following two types of applications, i.e., Taylor-type interest rate rules and government spending rules, given by

$$r_t = \alpha_r + \sum_{i=1}^{I} \rho_i r_{t-i} + \psi_1 \pi_t + \psi_2 ygap_t + \varepsilon_t, \tag{4.3}$$

and

$$g_t = \alpha_g + \sum_{i=1}^{I} \varrho_i g_{t-i} + \phi_1 b_{t-1} + \phi_2 ygap_t + \varepsilon_t, \tag{4.4}$$

where g_t denotes the government spending and b_t represents the government debt. Both series are scaled by GDP. To evaluate whether policy rules exhibit inertia, we allow I to range from zero to two. We consider three model specifications, each of which can be estimated using OLS and our proposed methods. This results in a total of nine models for comparison. The first set of models corresponds to the case of $I = 0$, meaning policy rules exclude lagged terms. These models, estimated using OLS, the one-step method, and the two-step method, are denoted as $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$, respectively. The second set of models allows for $I = 1$, incorporating a one-period lag in policy rules, with the corresponding estimates labeled $\mathcal{M}_4$, $\mathcal{M}_5$, and $\mathcal{M}_6$. Finally, the third set of models considers $I = 2$, where policy rules include a two-period lag, with the relevant estimates labeled $\mathcal{M}_7$, $\mathcal{M}_8$, and $\mathcal{M}_9$.

Each model is estimated using 10,000 draws of the MCMC algorithm. Specifically, we use the initial 15% of the total sample, spanning from 1987:Q4 to 2006:Q4, to form the prior. We sample 10,000 draws from the posterior distribution, discarding the first 1,000 as burn-in draws.

4.2 Tests of complete-information hypothesis We begin by testing the null hypothesis of complete information. If this null cannot be rejected, information rigidities are unlikely to generate additional predictability, and the one-step and two-step methods should deliver similar performance. By contrast, if the null is rejected, information rigidities are present, and the two-step methods are expected to outperform both our proposed one-step approach and OLS using real-time data. Specifically, we estimate the specification (2.5) for inflation and the output gap expectations, as shown in Table 2. The table reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th quantiles of the estimated parameters. A coefficient is considered statistically significant if the 2.5th and 97.5th percentiles share the same sign. Forecasts of inflation are calculated from PGDP in SPF and forecasts of the output gap are from Tealbook/Greenbook.

[14]Suppose that the prior probabilities are equal. Let δ_{lm} denote the difference in the log marginal likelihoods between models $\mathcal{M}_l$ and $\mathcal{M}_m$, which is calculated as $\ln m(y_{1:T}|\mathcal{M}_l, X_{1:T}) - \ln m(y_{1:T}|\mathcal{M}_m, X_{1:T})$. In a two-model case, the posterior probability of model l is greater than 67%, 75%, and 80%, which requires that δ_{lm} is greater than 0.694676, 1.098612, and 1.386294, respectively.

Over the sample period from 1987:Q4 to 2006:Q4, for both inflation and the output gap, the 2.5th and 97.5th percentiles exhibit opposite signs for coefficients on β. Therefore, we cannot reject the complete-information hypothesis. Based on the theoretical framework in Section 2, we expect that the one-step and two-step methods should have similar performance.

Table 2: Tests of inflation and output gap expectations processes

Inflation				Output gap			
Name	postmean	postsd	[0.025 0.975]	Name	postmean	postsd	[0.025 0.975]
α_π	−0.227	0.090	[−0.402 −0.051]	α_y	0.028	0.066	[−0.102 0.157]
β_π	−0.506	0.275	[−1.039 0.044]	β_y	−0.078	0.098	[−0.269 0.116]
σ_π^2	0.518	0.092	[0.367 0.725]	σ_y^2	0.303	0.054	[0.215 0.425]

Notes: The table reports coefficient estimates for specification (2.5) with respect to expectations of inflation and the output gap. We report the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Over the sample period from 1987:Q4 to 2006:Q4, for both inflation and the output gap, the 2.5th and 97.5th percentiles exhibit opposite signs for coefficients on β. Therefore, we cannot reject the complete-information hypothesis. Inflation forecasts are constructed from the SPF using the annualized quarter-over-quarter change in the GDP implicit price deflator (PGDP), and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Each model is estimated using 10,000 MCMC draws.

4.3 Monetary Policy Rule Estimations We start by estimating Taylor rules without policy inertia. Models $\mathcal{M}_1$–$\mathcal{M}_3$ represent the specification (4.3) when $I = 0$ using OLS with real-time data, our proposed one-step method and two-step method, respectively. Table 3 reports the posterior mean, posterior standard deviation (postsd), the 2.5th and 97.5th quantiles of the estimated parameters, and the log marginal likelihood for each specification. Our findings reveal that both the one-step and two-step methods yield higher marginal likelihoods than OLS estimates under all specifications. Since the null hypothesis of complete information over the sample period from 1987:Q4 to 2006:Q4 cannot be rejected, the one-step method and the two-step method exhibit similar performance across all specifications.

We extend our test to specifications with policy inertia. Previous studies on the estimation of monetary policy rules have highlighted the presence of policy inertia. Table 4 reports the estimates when the monetary policy rule has a one-period lag, i.e., $I = 1$ in specification (4.3). Our results confirm that our proposed one-step and two-step methods outperform estimating using real-time data, yielding larger log marginal likelihoods. In our empirical analysis, the coefficients on the one-period lagged federal funds rate are substantial (0.841, 0.717, and 0.717), with the 2.5th and 97.5th percentiles significantly different from zero across all specifications. This suggests that interest rate smoothing significantly improves the log marginal likelihoods for the OLS estimation, the one-step method, and the two-step method, raising the values from −122.9858, −95.39427, and −95.39427 to −43.86913, −34.35861, and −34.35856, respectively. Table 5 reports the estimates when the

monetary policy rule has two-period lags, i.e., $I = 2$ in the specification (4.3). The coefficients on the second-order interest rate in $\mathcal{M}_7$, $\mathcal{M}_8$, and $\mathcal{M}_9$ are negative and significantly below zero. Our results show that, over the sample period, our proposed method with a one-period interest rate lag yields the highest log marginal likelihood. These findings align with the conclusions in Coibion and Gorodnichenko (2012) and Woodford (2003b), suggesting that the central bank engaged in interest rate smoothing during the sample period.

For robustness, we consider several extensions. First, we re-estimate our models using SPF forecasts of core CPI and core PCE inflation. Second, to obtain a stricter real-time measure of the output gap, we construct gaps using real-time GDP from RTDSM and potential GDP from the CBO, rather than relying solely on Tealbook gaps, whose forecasts—while informative—are released with a five-year lag. Across these exercises, our proposed methods continue to outperform real-time OLS based on contemporaneous data. The corresponding estimates are reported in Appendix B.

4.4 Fiscal Policy Rule Estimations We estimate government spending rules in the specification (4.4). Since the null hypothesis of complete information over the sample period from 1987:Q4 to 2019:Q4 cannot be rejected, as shown in Table A.1, the one-step method and the two-step method should exhibit similar performance across all specifications.

We start by estimating government spending rules without policy inertia. Models $\mathcal{M}_1$–$\mathcal{M}_3$ represent the specification (4.3) when $I = 0$ using OLS estimation with real-time data, our proposed one-step method and two-step method, respectively. Table A.2 reports the posterior mean, posterior standard deviation (postsd), 2.5th and 97.5th quantiles of the estimated parameters, and log marginal likelihood for each specification. Our findings reveal that, for both debt and the output gap, the 2.5th and 97.5th percentiles exhibit the same signs for coefficients on ϕ_1 and ϕ_2. Both the one-step and two-step methods yield higher marginal likelihoods than OLS estimates under all specifications. Since the null hypothesis of complete information over the sample period from 1987:Q4 to 2019:Q4 cannot be rejected, the one-step method and the two-step method exhibit similar performance across all specifications.

Tables A.3 and A.4 compare OLS estimates based on real-time data with the one-step and two-step survey-based methods for specifications with one-period and two-period policy lags, respectively. The log marginal likelihoods are very similar across the three methods, which is consistent with the fact that the posterior 2.5th and 97.5th percentiles of the output-gap coefficient have opposite signs, indicating insignificance. Although forecasts of the output gap are informative, the insignificance of the corresponding coefficients suggests that government spending rules with policy lags do not necessarily respond to the output gap. Therefore, the forecasts cannot improve their performance due to the insignificance of the estimates on the output gap.

Table 3: Estimates of monetary policy without policy inertia

$\mathcal{M}_1$	postmean	postsd	[0.025 0.975]
α	2.248	0.351	[1.541 2.928]
$\psi_1 : \pi_t$	1.141	0.132	[0.890 1.405]
$\psi_2 : y_t$	0.641	0.085	[0.476 0.811]
σ^2	1.841	0.332	[1.299 2.592]
Log marginal likelihood	−122.9858		
$\mathcal{M}_2$	postmean	postsd	[0.025 0.975]
α	0.316	0.309	[−0.298 0.920]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	1.859	0.114	[1.636 2.083]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.696	0.056	[0.586 0.807]
σ^2	0.808	0.142	[0.576 1.126]
Log marginal likelihood	−95.39427		
$\mathcal{M}_3$	postmean	postsd	[0.025 0.975]
α	0.799	0.279	[0.242 1.345]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	3.112	0.192	[2.738 3.488]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.762	0.061	[0.642 0.883]
σ^2	0.808	0.142	[0.577 1.126]
Log marginal likelihood	−95.39427		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 0$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws.

Table 4: Estimates of monetary policy with a one-period interest rate lag

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.359	0.143	[0.082 0.643]
$\rho_1 : R_{t-1}$	0.841	0.033	[0.776 0.906]
$\psi_1 : \pi_t$	0.175	0.057	[0.065 0.288]
$\psi_2 : y_t$	0.173	0.031	[0.113 0.235]
$\psi_1^* : (1-\rho_1)\pi_t$	1.108	0.328	[0.488 1.794]
$\psi_2^* : (1-\rho_1)y_t$	1.119	0.229	[0.763 1.650]
σ^2	0.164	0.030	[0.115 0.231]
Log marginal likelihood	−43.86913		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.049	0.143	[−0.228 0.334]
$\rho_1 : R_{t-1}$	0.717	0.038	[0.643 0.790]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	0.540	0.089	[0.367 0.718]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.254	0.033	[0.191 0.317]
$\psi_1^* : (1-\rho_1)\mathbb{E}_{t-1}\pi_t$	1.912	0.209	[1.502 2.337]
$\psi_2^* : (1-\rho_1)\mathbb{E}_{t-1}y_t$	0.902	0.092	[0.741 1.102]
σ^2	0.126	0.023	[0.089 0.177]
Log marginal likelihood	−34.35861		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.187	0.131	[−0.068 0.450]
$\rho_1 : R_{t-1}$	0.717	0.038	[0.643 0.791]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	0.904	0.149	[0.616 1.200]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.278	0.036	[0.208 0.347]
$\psi_1^* : (1-\rho_1)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	3.201	0.349	[2.514 3.903]
$\psi_2^* : (1-\rho_1)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	0.987	0.101	[0.810 1.205]
σ^2	0.126	0.023	[0.089 0.177]
Log marginal likelihood	−34.35856		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1)$.

Table 5: Estimates of monetary policy with two-period interest rate lags

$\mathcal{M}_7$	postmean	postsd	[0.025 0.975]
α	0.319	0.119	[0.084 0.548]
$\rho_1 : R_{t-1}$	1.325	0.100	[1.129 1.519]
$\rho_2 : R_{t-2}$	−0.456	0.091	[−0.632 −0.276]
$\psi_1 : \pi_t$	0.134	0.049	[0.040 0.231]
$\psi_2 : y_t$	0.115	0.029	[0.059 0.171]
$\psi_1^* : (1-\rho_1-\rho_2)\pi_t$	1.027	0.330	[0.390 1.695]
$\psi_2^* : (1-\rho_1-\rho_2)y_t$	0.896	0.217	[0.527 1.385]
σ^2	0.121	0.022	[0.085 0.171]
Log marginal likelihood	−35.78617		
$\mathcal{M}_8$	postmean	postsd	[0.025 0.975]
α	0.065	0.117	[−0.164 0.291]
$\rho_1 : R_{t-1}$	1.044	0.118	[0.809 1.271]
$\rho_2 : R_{t-2}$	−0.282	0.095	[−0.464 −0.094]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	0.449	0.086	[0.282 0.619]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.198	0.036	[0.127 0.268]
$\psi_1^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}\pi_t$	1.879	0.190	[1.501 2.258]
$\psi_2^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}y_t$	0.832	0.100	[0.641 1.042]
σ^2	0.111	0.020	[0.078 0.156]
Log marginal likelihood	−35.57122		
$\mathcal{M}_9$	postmean	postsd	[0.025 0.975]
α	0.180	0.108	[−0.032 0.389]
$\rho_1 : R_{t-1}$	1.044	0.117	[0.809 1.269]
$\rho_2 : R_{t-2}$	−0.282	0.095	[−0.464 −0.095]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	0.752	0.144	[0.474 1.036]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.217	0.040	[0.140 0.294]
$\psi_1^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	3.146	0.318	[2.515 3.779]
$\psi_2^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	0.911	0.110	[0.702 1.139]
σ^2	0.111	0.020	[0.078 0.157]
Log marginal likelihood	−35.57120		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws.

The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1-\rho_2)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1-\rho_2)$.

5 Conclusion

This paper introduces empirical methods leveraging survey data to mitigate endogeneity bias in policy rule estimation across various information environments. We demonstrate that under complete information, our approach effectively eliminates endogeneity bias, while under incomplete information, we further orthogonalize forecast errors to address this bias. Our theoretical analysis establishes that these survey-based methods substantially reduce estimation bias in both information settings.

To quantify the magnitude of endogeneity bias, we first employ a standard new Keynesian model as our analytical framework. Our simulations reveal that conventional OLS methods can produce significantly inaccurate estimates, while incorporating survey data markedly improves estimation precision. We extend our analysis to a sticky information model, where simulations confirm that traditional OLS approaches yield biased outcomes. However, both our one-step and two-step methods substantially enhance estimation accuracy, with the two-step method further refining precision and bringing estimated policy coefficients closer to their true values.

We apply our proposed methodology to two key policy rules: interest rate and government spending rules. Our empirical results demonstrate that survey data significantly reduces endogeneity bias across both policy frameworks. Importantly, during our sample period, we cannot reject the null hypothesis of complete information, suggesting that information rigidities do not present a serious concern. This finding indicates that the one-step method is sufficient for our empirical purposes. Across nearly all model specifications, our survey-based approaches yield higher marginal likelihoods compared to standard OLS estimation, with the one-step method exhibits similar performance as the two-step method in most specifications.

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A Data Description

Data are from the Bureau of Economic Analysis (BEA), Survey of Professional Forecasters (SPF), Tealbook (formerly Greenbook) Data Set, and rchival Federal Reserve Economic Data (ALFRED), St. Louis Fed, Philadelphia Fed Real-Time Data Set for Macroeconomists (RTDSM), Congressional Budget Office (CBO).

In our main analysis, we consider monetary policy ranges from 1987:Q4 to 2006:Q4. For robustness check, we also consider time period from 2016 to 2019. To mitigate concerns that the post-2016 sample is too short at the quarterly frequency for a Cholesky-based decomposition, we convert all series to monthly frequency. Specifically, we take the monthly effective federal funds rate from FRED (St. Louis Fed) and construct monthly inflation and inflation-forecast series by assigning each quarterly observation to the three months of the corresponding quarter, thereby preserving the information set while increasing the number of time observations. We also consider fiscal policy ranges from 1987:Q4 to 2019:Q4.

1. **Nominal interest rate**: The policy rate is the effective federal funds rate (FEDFUNDS), obtained from the Federal Reserve Bank of St. Louis at a monthly frequency. To construct a quarterly series and mitigate serial correlation, we use the end-of-month observation from the second month of each quarter (February, May, August, and November).

2. **Output Gap**: Real-time output gaps and their forecasts are taken from the Tealbook/Greenbook. This data set provides the Federal Reserve Board staff's real-time estimates and projections of the output gap. The quarterly series are organized by vintage, and for each vintage the estimated and projected paths are based solely on information available at that vintage date.

   For a robustness check, we also construct a real-time output-gap series from RTDSM and potential real GDP is estimated by CBO and vintage data is downloaded from ALFRED. CBO updates potential output less frequently than BEA GDP vintages. For each calendar quarter, we take the first-available CBO potential GDP and convert its vintage release date into a quarter-level vintage index. We then align real GDP from RTDSM to this CBO vintage by selecting the RTDSM observation whose vintage quarter matches the CBO vintage quarter.

3. **Forecast of Inflation rate**: We calculated the annualized quarter-over-quarter forcasts of inflation rate using the mean PGDP in the Survey of Professional Forecasters (SPF).

   We use PGDP rather than core CPI or core PCE in the main analysis due to data availability. In the SPF, mean forecasts of core CPI and core PCE inflation are only available from 2007:Q1 onward, whereas our baseline sample spans 1987:Q4–2006:Q4. Instead, we employ core CPI and core PCE as robustness checks over the post-ZLB, pre-pandemic period from January 2015 to December 2019.

We use one-quarter-ahead, quarter-over-quarter inflation forecasts rather than year-over-year measures for two reasons. First, this construction ensures that the two PGDP observations entering the inflation calculation are conditioned on the same information set. Specifically, PGDP1 denotes the SPF "backcast" for the quarter preceding the survey quarter, and PGDP2 denotes the SPF forecast for the survey quarter itself. Both are formed at the same survey date, before the true value of contemporaneous PGDP is realized, so the implied inflation rate between these two quarters is a forecast made at time t based on information available at $t-1$. Second, although one-year-ahead year-over-year forecasts can be constructed using the same information set, four-quarter-ahead projections can be noisier due to long forecast horizons with less information. We therefore focus on one-quarter-ahead forecasts.

4. **Inflation rate**: We measure inflation as the annualized quarter-over-quarter change in the GDP implicit price deflator (PGDP). Real-time quarterly PGDP (GDPDEF) is obtained from BEA via ALFRED. For data prior to June 1, 1996, values are taken from the December 3, 1996, vintage. Starting from July 1, 1996, we select for each quarter the latest vintage released on or before that quarter's information date, thereby preserving a strictly real-time information set.

   As a robustness check, we replicate our key results using an inflation measure constructed from core CPI (BLS) and core PCE (BEA) levels obtained from ALFRED.

5. **Government spending**: Government spending is government consumption expenditures and gross investment (GCE) from BEA and vintage data is downloaded from ALFRED.

6. **Government debt**: Government spending is federal debt held by the public (FYGFDPUN) from BEA and vintage data is downloaded from ALFRED.

   Both government spending and government debt are scaled by GDP, which also from BEA via ALFRED. To align the vintages, for each observation date, the real-time values of GDP, GCE, and FYGFDPUN are taken from the earliest vintage month that is no earlier than the observation month and simultaneously available for all three series. Vintages are matched at the year–month level (ignoring the exact release day), and, when multiple vintages exist within that month, the earliest one is used.

B Robustness checks

In this section, our robustness checks include: (i) a post-ZLB, pre-pandemic window (2016–2019); (ii) replication with core PCE and core CPI where SPF forecasts are available; and (iii) alternative output-gap constructions using RTDSM GDP with CBO potential output.

We start with the 2016–2019 sample window. Table A.5 reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th quantiles of the estimated parameters. Over

the sample period from 2016 to 2019, the 2.5th and 97.5th percentiles exhibit the same signs for coefficients on inflation, while while exhibiting opposite signs for the output gap. Therefore, we cannot reject the complete-information hypothesis for the output gap. Forecasts of inflation are constructed from PGDP in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook).

Tables A.6–A.8 compare OLS estimates using real-time data with the one-step and two-step methods using survey data across various specifications of monetary policy rules in (4.3). Our findings demonstrate that the one-step and two-step methods yield higher marginal likelihoods than OLS estimations for all specifications over the sample period from 2016 to 2019. These results confirm that our proposed methods substantially reduce endogeneity.

Then, we turn to replications using core PCE and core CPI. Tables A.9 and A.10 report tests of the inflation and output-gap expectation processes over the 2016–2019 sample period, respectively. The results indicate that we cannot reject the complete-information hypothesis for either inflation or the output gap when using core CPI, nor for the output gap when using core PCE. Tables A.11–A.13 and A.14–A.16 compare OLS estimates based on real-time data with the one-step and two-step survey-based methods across various specifications of the monetary policy rule in (4.3), using core PCE and core CPI, respectively. Our findings confirm that the one-step and two-step methods yield higher marginal likelihoods than the OLS estimates for all specifications. Finally, we consider alternative output-gap constructions using RTDSM GDP together with CBO potential output. The results, reported in Tables A.17–A.19, further reinforce our main conclusions.

Table A.1: Tests of output-gap expectations process: 1987:Q4–2019:Q4

Name	postmean	postsd	[0.025	0.975]
α_y	0.021	0.037	[−0.053	0.093]
β_y	−0.031	0.063	[−0.155	0.096]
σ_y^2	0.210	0.027	[0.163	0.270]

Notes: The table reports coefficient estimates for the specification (2.5) with respect to expectations of the output gap. We report the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Over the sample period from 1987:Q4 to 2006:Q4, the 2.5th and 97.5th percentiles of the posterior distribution for β_y have opposite signs. Therefore, we cannot reject the complete-information hypothesis for output-gap expectations. The output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Each model is estimated using 10,000 MCMC draws.

Table A.2: Estimates of fiscal policy without policy inertia

$\mathcal{M}_1$	postmean	postsd	[0.025 0.975]
α	0.213	0.003	[0.207 0.219]
$\phi_1 : b_{t-1}$	−0.046	0.005	[−0.057 − 0.037]
$\phi_2 : y_t$	−0.003	0.000	[−0.003 − 0.002]
σ^2	0.00006	0.00001	[0.00004 0.00010]
Log marginal likelihood	322.828		
$\mathcal{M}_2$	postmean	postsd	[0.025 0.975]
α	0.205	0.002	[0.201 0.210]
$\phi_1 : b_{t-1}$	−0.036	0.004	[−0.044 − 0.029]
$\phi_2 : \mathbb{E}_{t-1} y_t$	−0.003	0.000	[−0.003 − 0.003]
σ^2	0.00004	0.00001	[0.00003 0.00005]
Log marginal likelihood	343.530		
$\mathcal{M}_3$	postmean	postsd	[0.025 0.975]
α	0.205	0.002	[0.201 0.210]
$\phi_1 : b_{t-1}$	−0.036	0.004	[−0.044 − 0.029]
$\phi_2 : \mathbb{E}_{t-1} y_t + \widehat{\Delta y}_{t,t-1}$	−0.003	0.000	[−0.004 − 0.003]
σ^2	0.00004	0.00001	[0.00003 0.00005]
Log marginal likelihood	343.530		

Notes: The table reports posterior estimates for the rule in the specification (4.4) without policy inertia. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of each parameter. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based specifications, $\mathcal{M}_2$ and $\mathcal{M}_3$, achieve substantially higher log marginal likelihoods than the real-time benchmark $\mathcal{M}_1$, indicating superior fit. The sample covers 1987:Q4–2019:Q4. Each model is estimated using 10,000 MCMC draws.

Table A.3: Estimates of fiscal policy with a one-period lag

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.013	0.005	[0.002 0.024]
$\varrho_1 : g_{t-1}$	0.943	0.028	[0.888 0.997]
$\phi_1 : b_{t-1}$	−0.005	0.001	[−0.007 −0.002]
$\phi_2 : y_t$	-1.038×10^{-4}	1.090×10^{-4}	$[-3.204 \times 10^{-4}\ \ 1.096 \times 10^{-4}]$
$\phi_1^* : (1-\varrho_1)b_{t-1}$	−0.046	6.102	[−0.375 −0.034]
$\phi_2^* : (1-\varrho_1)y_t$	−0.0007	0.0463	[−0.0046 0.0068]
σ^2	2.886×10^{-6}	3.956×10^{-7}	$[2.212 \times 10^{-6}\ \ 3.767 \times 10^{-6}]$
Log marginal likelihood	533.877		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.010	0.006	[−0.001 0.021]
$\varrho_1 : g_{t-1}$	0.958	0.028	[0.901 1.014]
$\phi_1 : b_{t-1}$	−0.004	0.001	[−0.007 −0.002]
$\phi_2 : \mathbb{E}_{t-1}y_t$	-3.294×10^{-5}	1.117×10^{-4}	$[-2.550 \times 10^{-4}\ \ 1.856 \times 10^{-4}]$
$\phi_1^* : (1-\varrho_1)b_{t-1}$	−0.036	18.004	[−0.723 0.485]
$\phi_2^* : (1-\varrho_1)\mathbb{E}_{t-1}y_t$	−0.004	0.747	[−0.019 0.020]
σ^2	2.923×10^{-6}	4.006×10^{-7}	$[2.241 \times 10^{-6}\ \ 3.817 \times 10^{-6}]$
Log marginal likelihood	533.817		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.010	0.006	[−0.001 0.021]
$\varrho_1 : g_{t-1}$	0.958	0.028	[0.901 1.014]
$\phi_1 : b_{t-1}$	−0.004	0.001	[−0.007 −0.002]
$\phi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	-3.456×10^{-5}	1.172×10^{-4}	$[-2.675 \times 10^{-4}\ \ 1.949 \times 10^{-4}]$
$\phi_1^* : (1-\varrho_1)b_{t-1}$	−0.051	16.294	[−0.723 0.485]
$\phi_2^* : (1-\varrho_1)(\mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1})$	−0.003	0.710	[−0.020 0.020]
σ^2	2.923×10^{-6}	4.006×10^{-7}	$[2.241 \times 10^{-6}\ \ 3.817 \times 10^{-6}]$
Log marginal likelihood	533.817		

Notes: The table reports coefficient estimates for the specification (4.4) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. The three methods yield similar performance as the coefficients on the output gap are insignificant. The sample covers 1987:Q4–2019:Q4. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\phi_1^* \equiv \phi_1/(1-\varrho_1)$ and $\phi_2^* \equiv \phi_2/(1-\varrho_1)$.

Table A.4: Estimates of fiscal policy with two-period lags

$\mathcal{M}_7$	postmean	postsd	[0.025	0.975]
α	1.243×10^{-2}	5.230×10^{-3}	$[2.153 \times 10^{-3}$	$2.255 \times 10^{-2}]$
$\varrho_1 : g_{t-1}$	1.184	0.092	[1.002	1.363]
$\varrho_2 : g_{t-2}$	-0.241	0.089	$[-0.413$	$-0.066]$
$\phi_1 : b_{t-1}$	-0.0038	0.0012	$[-0.0061$	$-0.0014]$
$\phi_2 : y_t$	-1.139×10^{-4}	1.048×10^{-4}	$[-3.195 \times 10^{-4}$	$9.325 \times 10^{-5}]$
$\phi_1^* : (1-\varrho_1-\varrho_2)b_{t-1}$	-0.086	1.299	$[-0.270$	$-0.023]$
$\phi_2^* : (1-\varrho_1-\varrho_2)y_t$	-0.0011	0.084	$[-0.0047$	$0.0050]$
σ^2	2.733×10^{-6}	3.754×10^{-7}	$[2.093 \times 10^{-6}$	$3.570 \times 10^{-6}]$
Log marginal likelihood	535.607			
$\mathcal{M}_8$	postmean	postsd	[0.025	0.975]
α	1.037×10^{-2}	5.336×10^{-3}	$[-1.008 \times 10^{-4}$	$2.069 \times 10^{-2}]$
$\varrho_1 : g_{t-1}$	1.196	0.092	[1.016	1.375]
$\varrho_2 : g_{t-2}$	-0.243	0.090	$[-0.417$	$-0.066]$
$\phi_1 : b_{t-1}$	-0.0035	0.0012	$[-0.0058$	$-0.0011]$
$\phi_2 : \mathbb{E}_{t-1}y_t$	-6.334×10^{-5}	1.069×10^{-4}	$[-2.727 \times 10^{-4}$	$1.472 \times 10^{-4}]$
$\phi_1^* : (1-\varrho_1-\varrho_2)b_{t-1}$	-0.127	3.194	$[-0.449$	$0.189]$
$\phi_2^* : (1-\varrho_1-\varrho_2)\mathbb{E}_{t-1}y_t$	0.0015	0.131	$[-0.0114$	$0.0138]$
σ^2	2.764×10^{-6}	3.794×10^{-7}	$[2.117 \times 10^{-6}$	$3.607 \times 10^{-6}]$
Log marginal likelihood	535.397			
$\mathcal{M}_9$	postmean	postsd	[0.025	0.975]
α	1.037×10^{-2}	5.337×10^{-3}	$[-1.028 \times 10^{-4}$	$2.069 \times 10^{-2}]$
$\varrho_1 : g_{t-1}$	1.196	0.092	[1.016	1.375]
$\varrho_2 : g_{t-2}$	-0.243	0.090	$[-0.417$	$-0.066]$
$\phi_1 : b_{t-1}$	-0.0035	0.0012	$[-0.0058$	$-0.0011]$
$\phi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	-6.644×10^{-5}	1.122×10^{-4}	$[-2.863 \times 10^{-4}$	$1.541 \times 10^{-4}]$
$\phi_1^* : (1-\varrho_1-\varrho_2)b_{t-1}$	-0.129	3.165	$[-0.449$	$0.189]$
$\phi_2^* : (1-\varrho_1-\varrho_2)(\mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1})$	0.0016	0.140	$[-0.0120$	$0.0144]$
σ^2	2.764×10^{-6}	3.794×10^{-7}	$[2.117 \times 10^{-6}$	$3.607 \times 10^{-6}]$
Log marginal likelihood	535.397			

Notes: The table reports coefficient estimates for the specification (4.4) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. The three methods yield similar performance as the coefficients on the output gap are insignificant. The sample covers 1987:Q4–2019:Q4. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\phi_1^* \equiv \phi_1/(1-\varrho_1-\varrho_2)$ and $\phi_2^* \equiv \phi_2/(1-\varrho_1-\varrho_2)$.

Table A.5: Tests of inflation and output gap expectations processes, 2016-2019

Inflation					Output gap				
Name	postmean	postsd	[0.025	0.975]	Name	postmean	postsd	[0.025	0.975]
α_π	−0.297	0.117	[−0.535	− 0.077]	α_y	−0.071	0.029	[−0.129	− 0.014]
β_π	−0.836	0.396	[−1.621	− 0.069]	β_y	0.126	0.139	[−0.147	0.402]
σ^2_π	0.473	0.110	[0.304	0.728]	σ^2_y	0.038	0.009	[0.025	0.058]

Notes: The table reports coefficient estimates for the specification (2.5) with respect to expectations of inflation and the output gap. We report the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Over the sample period from 2016 to 2019, the 2.5th and 97.5th percentiles exhibit the same signs for coefficients on β_π, while exhibit opposite signs for β_y. Therefore, we cannot reject the complete-information hypothesis for the output gap. Inflation forecasts are constructed from the SPF using the annualized quarter-over-quarter change in the GDP implicit price deflator (PGDP), and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Each model is estimated using 10,000 MCMC draws.

Table A.6: Estimates of monetary policy without policy inertia, 2016-2019

$\mathcal{M}_1$	postmean	postsd	[0.025	0.975]
α	0.398	0.034	[0.331	0.464]
$\psi_1 : \pi_t$	−0.016	0.031	[−0.076	0.046]
$\psi_2 : y_t$	0.841	0.033	[0.774	0.905]
σ^2	0.073	0.017	[0.046	0.112]
Log marginal likelihood	−26.2615			
$\mathcal{M}_2$	postmean	postsd	[0.025	0.975]
α	0.497	0.067	[0.365	0.630]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.132	0.050	[−0.230	− 0.035]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.872	0.028	[0.818	0.927]
σ^2	0.037	0.008	[0.024	0.056]
Log marginal likelihood	−6.992376			
$\mathcal{M}_3$	postmean	postsd	[0.025	0.975]
α	0.508	0.077	[0.356	0.659]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	−0.122	0.046	[−0.213	− 0.032]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.915	0.029	[0.858	0.973]
σ^2	0.037	0.008	[0.024	0.056]
Log marginal likelihood	−6.992394			

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 0$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws.

Table A.7: Estimates of monetary policy with a one-period interest rate lag, 2016–2019

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.082	0.030	[0.024 0.142]
$\rho_1 : R_{t-1}$	0.811	0.045	[0.721 0.899]
$\psi_1 : \pi_t$	0.0069	0.0178	[−0.028 0.041]
$\psi_2 : y_t$	0.158	0.044	[0.073 0.246]
$\psi_1^* : (1-\rho_1)^{-1}\pi_t$	0.049	0.115	[−0.133 0.312]
$\psi_2^* : (1-\rho_1)^{-1}y_t$	0.828	0.086	[0.640 0.977]
σ^2	0.0077	0.0018	[0.0050 0.0118]
Log marginal likelihood	24.0963		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.172	0.055	[0.064 0.283]
$\rho_1 : R_{t-1}$	0.704	0.050	[0.603 0.801]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.044	0.037	[−0.119 0.028]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.260	0.046	[0.170 0.354]
$\psi_1^* : (1-\rho_1)^{-1}\mathbb{E}_{t-1}\pi_t$	−0.151	0.132	[−0.422 0.101]
$\psi_2^* : (1-\rho_1)^{-1}\mathbb{E}_{t-1}y_t$	0.878	0.055	[0.771 0.990]
σ^2	0.0063	0.0015	[0.0041 0.0097]
Log marginal likelihood	27.8574		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.176	0.062	[0.054 0.300]
$\rho_1 : R_{t-1}$	0.704	0.050	[0.603 0.801]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	−0.041	0.034	[−0.109 0.026]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.273	0.048	[0.179 0.371]
$\psi_1^* : (1-\rho_1)^{-1}\big(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}\big)$	−0.139	0.122	[−0.386 0.094]
$\psi_2^* : (1-\rho_1)^{-1}\big(\mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}\big)$	0.921	0.058	[0.809 1.039]
σ^2	0.0063	0.0015	[0.0041 0.0097]
Log marginal likelihood	27.8575		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected for the output gap. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1)$

Table A.8: Estimates of monetary policy with two-period interest rate lags, 2016–2019

$\mathcal{M}_7$	postmean	postsd	[0.025 0.975]
α	0.069	0.027	[0.016 0.122]
$\rho_1 : R_{t-1}$	1.182	0.124	[0.940 1.425]
$\rho_2 : R_{t-2}$	−0.340	0.107	[−0.550 − 0.132]
$\psi_1 : \pi_t$	−0.003	0.016	[−0.034 0.029]
$\psi_2 : y_t$	0.138	0.041	[0.058 0.218]
$\psi_1^* : (1-\rho_1-\rho_2)^{-1}\pi_t$	−0.007	0.176	[−0.221 0.255]
$\psi_2^* : (1-\rho_1-\rho_2)^{-1}y_t$	0.872	0.139	[0.665 1.058]
σ^2	0.0069	0.0016	[0.0045 0.0105]
Log marginal likelihood	25.3497		
$\mathcal{M}_8$	postmean	postsd	[0.025 0.975]
α	0.095	0.058	[−0.018 0.209]
$\rho_1 : R_{t-1}$	1.062	0.139	[0.795 1.336]
$\rho_2 : R_{t-2}$	−0.311	0.112	[−0.537 − 0.096]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.006	0.036	[−0.078 0.064]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.210	0.048	[0.116 0.306]
$\psi_1^* : (1-\rho_1-\rho_2)^{-1}\mathbb{E}_{t-1}\pi_t$	−0.017	0.158	[−0.305 0.317]
$\psi_2^* : (1-\rho_1-\rho_2)^{-1}\mathbb{E}_{t-1}y_t$	0.840	0.066	[0.703 0.962]
σ^2	0.0060	0.0014	[0.0039 0.0092]
Log marginal likelihood	28.3315		
$\mathcal{M}_9$	postmean	postsd	[0.025 0.975]
α	0.092	0.065	[−0.034 0.219]
$\rho_1 : R_{t-1}$	1.062	0.139	[0.796 1.337]
$\rho_2 : R_{t-2}$	−0.311	0.112	[−0.537 − 0.096]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	−0.006	0.034	[−0.072 0.060]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.220	0.051	[0.122 0.320]
$\psi_1^* : (1-\rho_1-\rho_2)^{-1}\big(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}\big)$	−0.016	0.146	[−0.282 0.293]
$\psi_2^* : (1-\rho_1-\rho_2)^{-1}\big(\mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}\big)$	0.881	0.069	[0.738 1.009]
σ^2	0.0060	0.0014	[0.0039 0.0092]
Log marginal likelihood	28.3315		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1-\rho_2)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1-\rho_2)$.

Table A.9: Tests of inflation (core PCE) and the output gap expectations processes, 2016–2019

Inflation				Output gap			
Name	postmean	postsd	[0.025 0.975]	Name	postmean	postsd	[0.025 0.975]
α_π	0.030	0.038	[−0.045 0.105]	α_y	−0.071	0.029	[−0.129 − 0.014]
β_π	2.246	0.286	[1.683 2.812]	β_y	0.126	0.139	[−0.147 0.402]
σ_π^2	0.227	0.055	[0.142 0.354]	σ_y^2	0.038	0.009	[0.025 0.058]

Notes: The table reports coefficient estimates for specification (2.5) with respect to expectations of inflation and the output gap. We report the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Over the sample period from 2016 to 2019, the 2.5th and 97.5th percentiles exhibit the same signs for coefficients on β_π, while opposite signs for β_y. Therefore, we cannot reject the complete-information hypothesis for the output gap. Inflation forecasts are constructed from core PCE in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Each model is estimated using 10,000 MCMC draws.

Table A.10: Tests of inflation (core CPI) and the output gap expectations processes, 2016–2019

Inflation				Output gap			
Name	postmean	postsd	[0.025 0.975]	Name	postmean	postsd	[0.025 0.975]
α_π	−0.117	0.222	[−0.550 0.325]	α_y	−0.071	0.029	[−0.129 − 0.014]
β_π	0.212	0.524	[−0.825 1.222]	β_y	0.126	0.139	[−0.147 0.402]
σ_π^2	0.331	0.080	[0.209 0.519]	σ_y^2	0.038	0.009	[0.025 0.058]

Notes: The table reports coefficient estimates for specification (2.5) with respect to expectations of inflation and the output gap. We report the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Over the sample period from 2016 to 2019, the 2.5th and 97.5th percentiles exhibit opposite signs for the coefficients on both β_π and β_y. Therefore, we cannot reject the complete-information hypothesis for either inflation or the output gap. Inflation forecasts are constructed from core CPI in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Each model is estimated using 10,000 MCMC draws.

Table A.11: Estimates of monetary policy without policy inertia: core PCE

$\mathcal{M}_1$	postmean	postsd	[0.025 0.975]
α	0.316	0.041	[0.237 0.396]
$\psi_1 : \pi_t$	0.048	0.029	[−0.008 0.104]
$\psi_2 : y_t$	0.831	0.029	[0.772 0.889]
σ^2	0.078	0.017	[0.051 0.118]
Log marginal likelihood	−24.91059		
$\mathcal{M}_2$	postmean	postsd	[0.025 0.975]
α	0.718	0.140	[0.443 0.993]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.268	0.095	[−0.453 − 0.083]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.883	0.029	[0.826 0.941]
σ^2	0.036	0.008	[0.024 0.055]
Log marginal likelihood	−6.790681		
$\mathcal{M}_3$	postmean	postsd	[0.025 0.975]
α	0.735	0.127	[0.485 0.985]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	−0.144	0.051	[−0.243 − 0.045]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.790	0.026	[0.738 0.842]
σ^2	0.036	0.008	[0.024 0.055]
Log marginal likelihood	−6.790638		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 0$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. The results indicate that all estimates are significant throughout the sample period. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from core PCE in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws.

Table A.12: Estimates of monetary policy with a one-period interest rate lag: core PCE

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.137	0.031	[0.076 0.199]
$\rho_1 : R_{t-1}$	0.807	0.040	[0.726 0.885]
$\psi_1 : \pi_t$	−0.034	0.020	[−0.072 0.004]
$\psi_2 : y_t$	0.168	0.036	[0.098 0.238]
$\psi_1^* : (1-\rho_1)\pi_t$	−0.185	0.118	[−0.441 0.021]
$\psi_2^* : (1-\rho_1)y_t$	0.869	0.056	[0.764 0.981]
σ^2	0.007	0.002	[0.005 0.011]
Log marginal likelihood	25.84345		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.292	0.102	[0.092 0.499]
$\rho_1 : R_{t-1}$	0.701	0.050	[0.602 0.797]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.120	0.067	[−0.252 0.009]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.272	0.046	[0.183 0.366]
$\psi_1^* : (1-\rho_1)\mathbb{E}_{t-1}\pi_t$	−0.410	0.236	[−0.898 0.032]
$\psi_2^* : (1-\rho_1)\mathbb{E}_{t-1}y_t$	0.912	0.058	[0.801 1.030]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	28.43934		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.292	0.094	[0.108 0.481]
$\rho_1 : R_{t-1}$	0.701	0.050	[0.602 0.798]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	−0.065	0.036	[−0.136 0.005]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.244	0.041	[0.163 0.327]
$\psi_1^* : (1-\rho_1)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	−0.220	0.127	[−0.483 0.016]
$\psi_2^* : (1-\rho_1)(\mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1})$	0.816	0.052	[0.717 0.921]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	28.43922		

Notes: The table reports coefficient estimates for specification (4.3) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are core PCE in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1)$.

Table A.13: Estimates of monetary policy with two-period interest rate lags: core PCE

$\mathcal{M}_7$	postmean	postsd	[0.025 0.975]
α	0.073	0.037	[−0.000 0.144]
$\rho_1 : R_{t-1}$	1.165	0.141	[0.893 1.448]
$\rho_2 : R_{t-2}$	−0.323	0.121	[−0.566 − 0.089]
$\psi_1 : \pi_t$	−0.005	0.020	[−0.044 0.036]
$\psi_2 : y_t$	0.136	0.036	[0.065 0.207]
$\psi_1^* : (1-\rho_1-\rho_2)\pi_t$	−0.026	0.165	[−0.304 0.281]
$\psi_2^* : (1-\rho_1-\rho_2)y_t$	0.867	0.094	[0.732 1.015]
σ^2	0.007	0.002	[0.004 0.011]
Log marginal likelihood	25.79455		
$\mathcal{M}_8$	postmean	postsd	[0.025 0.975]
α	0.169	0.108	[−0.041 0.380]
$\rho_1 : R_{t-1}$	1.019	0.139	[0.751 1.297]
$\rho_2 : R_{t-2}$	−0.277	0.112	[−0.502 − 0.061]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.053	0.067	[−0.185 0.078]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.226	0.049	[0.131 0.323]
$\psi_1^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}\pi_t$	−0.201	0.271	[−0.718 0.352]
$\psi_2^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}y_t$	0.875	0.067	[0.741 1.003]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	28.28389		
$\mathcal{M}_9$	postmean	postsd	[0.025 0.975]
α	0.176	0.100	[−0.019 0.372]
$\rho_1 : R_{t-1}$	1.019	0.139	[0.752 1.297]
$\rho_2 : R_{t-2}$	−0.277	0.112	[−0.503 − 0.061]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	−0.029	0.036	[−0.100 0.042]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.202	0.044	[0.116 0.289]
$\psi_1^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1})$	−0.108	0.146	[−0.386 0.188]
$\psi_2^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1})$	0.783	0.060	[0.662 0.897]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	28.28393		

Notes: The table reports coefficient estimates for specification (4.3) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from core PCE in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1-\rho_2)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1-\rho_2)$.

Table A.14: Estimates of monetary policy without policy inertia: core CPI

$\mathcal{M}_1$	postmean	postsd	[0.025 0.975]
α	0.208	0.073	[0.064 0.349]
$\psi_1 : \pi_t$	0.084	0.035	[0.016 0.153]
$\psi_2 : y_t$	0.851	0.031	[0.791 0.913]
σ^2	0.081	0.018	[0.053 0.123]
Log marginal likelihood	−23.86008		
$\mathcal{M}_2$	postmean	postsd	[0.025 0.975]
α	0.866	0.171	[0.531 1.200]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.289	0.091	[−0.467 − 0.111]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.869	0.024	[0.822 0.917]
σ^2	0.034	0.007	[0.022 0.051]
Log marginal likelihood	−5.059826		
$\mathcal{M}_3$	postmean	postsd	[0.025 0.975]
α	0.887	0.161	[0.572 1.201]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	−0.215	0.067	[−0.347 − 0.083]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.777	0.022	[0.735 0.820]
σ^2	0.034	0.007	[0.022 0.051]
Log marginal likelihood	−5.059806		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 0$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from core CPI in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws.

Table A.15: Estimates of monetary policy with a one-period interest rate lag: core CPI

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.126	0.040	[0.047 0.206]
$\rho_1 : R_{t-1}$	0.807	0.041	[0.725 0.887]
$\psi_1 : \pi_t$	−0.017	0.018	[−0.053 0.019]
$\psi_2 : y_t$	0.163	0.037	[0.092 0.236]
$\psi_1^* : (1-\rho_1)\pi_t$	−0.094	0.106	[−0.320 0.096]
$\psi_2^* : (1-\rho_1)y_t$	0.842	0.060	[0.726 0.957]
σ^2	0.008	0.002	[0.005 0.012]
Log marginal likelihood	24.01371		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.331	0.124	[0.088 0.580]
$\rho_1 : R_{t-1}$	0.690	0.051	[0.588 0.788]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.113	0.063	[−0.240 0.010]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.273	0.046	[0.183 0.367]
$\psi_1^* : (1-\rho_1)\mathbb{E}_{t-1}\pi_t$	−0.368	0.207	[−0.785 0.032]
$\psi_2^* : (1-\rho_1)\mathbb{E}_{t-1}y_t$	0.881	0.043	[0.797 0.966]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	29.12295		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.335	0.118	[0.104 0.572]
$\rho_1 : R_{t-1}$	0.690	0.051	[0.588 0.788]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1}$	−0.084	0.047	[−0.178 0.007]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1}$	0.244	0.041	[0.164 0.328]
$\psi_1^* : (1-\rho_1)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta\pi}_{t,t-1})$	−0.273	0.154	[−0.582 0.025]
$\psi_2^* : (1-\rho_1)(\mathbb{E}_{t-1}y_t + \widehat{\Delta y}_{t,t-1})$	0.788	0.038	[0.713 0.864]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	29.12285		

Notes: The table reports coefficient estimates for specification (4.3) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from core CPI in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1)$.

Table A.16: Estimates of monetary policy with two-period interest rate lags: core CPI

$\mathcal{M}_7$	postmean	postsd	[0.025 0.975]
α	0.062	0.043	[−0.022 0.144]
$\rho_1 : R_{t-1}$	1.184	0.130	[0.934 1.440]
$\rho_2 : R_{t-2}$	−0.340	0.111	[−0.562 −0.126]
$\psi_1 : \pi_t$	0.002	0.018	[−0.033 0.038]
$\psi_2 : y_t$	0.135	0.036	[0.063 0.206]
$\psi_1^* : (1-\rho_1-\rho_2)\pi_t$	0.016	0.163	[−0.238 0.294]
$\psi_2^* : (1-\rho_1-\rho_2)y_t$	0.868	0.091	[0.727 1.027]
σ^2	0.007	0.002	[0.004 0.011]
Log marginal likelihood	24.92798		
$\mathcal{M}_8$	postmean	postsd	[0.025 0.975]
α	0.203	0.126	[−0.044 0.451]
$\rho_1 : R_{t-1}$	1.012	0.138	[0.745 1.284]
$\rho_2 : R_{t-2}$	−0.276	0.108	[−0.495 −0.066]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	−0.059	0.062	[−0.180 0.063]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.229	0.049	[0.134 0.324]
$\psi_1^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}\pi_t$	−0.213	0.241	[−0.663 0.288]
$\psi_2^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}y_t$	0.864	0.050	[0.763 0.960]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	29.08638		
$\mathcal{M}_9$	postmean	postsd	[0.025 0.975]
α	0.210	0.121	[−0.025 0.447]
$\rho_1 : R_{t-1}$	1.012	0.138	[0.746 1.285]
$\rho_2 : R_{t-2}$	−0.276	0.108	[−0.494 −0.066]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	−0.044	0.046	[−0.135 0.047]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.204	0.043	[0.120 0.290]
$\psi_1^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1})$	−0.159	0.179	[−0.494 0.213]
$\psi_2^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1})$	0.773	0.044	[0.683 0.859]
σ^2	0.006	0.001	[0.004 0.009]
Log marginal likelihood	29.08641		

Notes: The table reports coefficient estimates for specification (4.3) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and the 2.5th and 97.5th percentiles of the posterior distribution for each parameter. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from core CPI in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 2016–2019. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1-\rho_2)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1-\rho_2)$.

Table A.17: Estimates of monetary policy without policy inertia: RTDSM

$\mathcal{M}_1$	postmean	postsd	[0.025 0.975]
α	2.066	0.444	[1.171 2.923]
$\psi_1 : \pi_t$	1.125	0.175	[0.793 1.476]
$\psi_2 : y_t$	0.340	0.085	[0.175 0.510]
σ^2	2.662	0.480	[1.880 3.740]
Log marginal likelihood	−133.5303		
$\mathcal{M}_2$	postmean	postsd	[0.025 0.975]
α	0.316	0.309	[−0.298 0.920]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	1.859	0.114	[1.636 2.083]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.696	0.056	[0.586 0.807]
σ^2	0.808	0.142	[0.576 1.126]
Log marginal likelihood	−95.39427		
$\mathcal{M}_3$	postmean	postsd	[0.025 0.975]
α	0.799	0.279	[0.242 1.345]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	3.112	0.192	[2.738 3.488]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.762	0.061	[0.642 0.883]
σ^2	0.808	0.142	[0.577 1.126]
Log marginal likelihood	−95.39427		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 0$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. Models $\mathcal{M}_1$–$\mathcal{M}_3$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from PGDP in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). The real-time output-gap series is from RTDSM and the potential output gap is estimated by the CBO. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws.

Table A.18: Estimates of monetary policy with a one-period interest rate lag: RTDSM

$\mathcal{M}_4$	postmean	postsd	[0.025 0.975]
α	0.069	0.137	[−0.197 0.341]
$\rho_1 : R_{t-1}$	0.898	0.031	[0.836 0.959]
$\psi_1 : \pi_t$	0.172	0.057	[0.063 0.285]
$\psi_2 : y_t$	0.097	0.025	[0.048 0.147]
$\psi_1^* : (1-\rho_1)\pi_t$	1.754	1.430	[0.811 3.177]
$\psi_2^* : (1-\rho_1)y_t$	1.030	1.122	[0.494 2.126]
σ^2	0.212	0.039	[0.149 0.300]
Log marginal likelihood	−54.49337		
$\mathcal{M}_5$	postmean	postsd	[0.025 0.975]
α	0.049	0.143	[−0.228 0.334]
$\rho_1 : R_{t-1}$	0.717	0.038	[0.643 0.790]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	0.540	0.089	[0.367 0.718]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.254	0.033	[0.191 0.317]
$\psi_1^* : (1-\rho_1)\mathbb{E}_{t-1}\pi_t$	1.912	0.209	[1.502 2.337]
$\psi_2^* : (1-\rho_1)\mathbb{E}_{t-1}y_t$	0.902	0.092	[0.741 1.102]
σ^2	0.126	0.023	[0.089 0.177]
Log marginal likelihood	−34.35861		
$\mathcal{M}_6$	postmean	postsd	[0.025 0.975]
α	0.187	0.131	[−0.068 0.450]
$\rho_1 : R_{t-1}$	0.717	0.038	[0.643 0.791]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	0.904	0.149	[0.616 1.200]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.278	0.036	[0.208 0.347]
$\psi_1^* : (1-\rho_1)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1})$	3.201	0.349	[2.514 3.903]
$\psi_2^* : (1-\rho_1)(\mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1})$	0.987	0.101	[0.810 1.205]
σ^2	0.126	0.023	[0.089 0.177]
Log marginal likelihood	−34.35856		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 1$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. Models $\mathcal{M}_4$–$\mathcal{M}_6$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from PGDP in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). The real-time output-gap series is from RTDSM and the potential output gap is estimated by the CBO. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1)$.

Table A.19: Estimates of monetary policy with two-period interest rate lags: RTDSM

$\mathcal{M}_7$	postmean	postsd	[0.025 0.975]
α	0.146	0.116	[−0.082 0.369]
$\rho_1 : R_{t-1}$	1.444	0.097	[1.249 1.631]
$\rho_2 : R_{t-2}$	−0.533	0.092	[−0.710 − 0.348]
$\psi_1 : \pi_t$	0.113	0.049	[0.020 0.209]
$\psi_2 : y_t$	0.051	0.022	[0.008 0.094]
$\psi_1^* : (1-\rho_1-\rho_2)\pi_t$	1.305	0.680	[0.315 2.456]
$\psi_2^* : (1-\rho_1-\rho_2)y_t$	0.616	0.520	[0.108 1.318]
σ^2	0.142	0.026	[0.100 0.202]
Log marginal likelihood	−42.40841		
$\mathcal{M}_8$	postmean	postsd	[0.025 0.975]
α	0.065	0.117	[−0.164 0.291]
$\rho_1 : R_{t-1}$	1.044	0.118	[0.809 1.271]
$\rho_2 : R_{t-2}$	−0.282	0.095	[−0.464 − 0.094]
$\psi_1 : \mathbb{E}_{t-1}\pi_t$	0.449	0.086	[0.282 0.619]
$\psi_2 : \mathbb{E}_{t-1}y_t$	0.198	0.036	[0.127 0.268]
$\psi_1^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}\pi_t$	1.879	0.190	[1.501 2.258]
$\psi_2^* : (1-\rho_1-\rho_2)\mathbb{E}_{t-1}y_t$	0.832	0.100	[0.641 1.042]
σ^2	0.111	0.020	[0.078 0.156]
Log marginal likelihood	−35.57122		
$\mathcal{M}_9$	postmean	postsd	[0.025 0.975]
α	0.180	0.108	[−0.032 0.389]
$\rho_1 : R_{t-1}$	1.044	0.117	[0.809 1.269]
$\rho_2 : R_{t-2}$	−0.282	0.095	[−0.464 − 0.095]
$\psi_1 : \mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1}$	0.752	0.144	[0.474 1.036]
$\psi_2 : \mathbb{E}_{t-1}y_t + \widehat{\Delta}y_{t,t-1}$	0.217	0.040	[0.140 0.294]
$\psi_1^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1})$	3.146	0.318	[2.515 3.779]
$\psi_2^* : (1-\rho_1-\rho_2)(\mathbb{E}_{t-1}\pi_t + \widehat{\Delta}\pi_{t,t-1})$	0.911	0.110	[0.702 1.139]
σ^2	0.111	0.020	[0.078 0.157]
Log marginal likelihood	−35.57120		

Notes: The table reports coefficient estimates for the specification (4.3) with $I = 2$. It reports the posterior mean, posterior standard deviation (postsd), and 2.5th and 97.5th percentiles of the estimated parameters. Models $\mathcal{M}_7$–$\mathcal{M}_9$ correspond to estimates obtained using OLS with real-time data, the one-step method with survey data, and the two-step method with survey data, respectively. Inflation forecasts are constructed from PGDP in the SPF, and output-gap forecasts are taken from the Tealbook (formerly the Greenbook). The real-time output-gap series is from RTDSM and the potential output gap is estimated by the CBO. Our survey-based methods outperform real-time OLS, as indicated by higher log marginal likelihoods. One-step and two-step procedures perform similarly, consistent with the complete-information hypothesis not being rejected. The sample covers 1987:Q4–2006:Q4. Each model is estimated using 10,000 MCMC draws. The rescaled parameters are defined as $\psi_1^* \equiv \psi_1/(1-\rho_1-\rho_2)$ and $\psi_2^* \equiv \psi_2/(1-\rho_1-\rho_2)$.

C MONETARY POLICY SHOCK VARIANCE ON ESTIMATION

Carvalho et al. (2021) show that monetary policy shock variance affects estimation performance. We first verify that shocks are not excessively large in our baseline calibration. Carvalho et al. (2021) calibrate $\sigma_R/\sigma_\pi = 0.45$ to match estimates from Smets and Wouters (2007). We use this as a benchmark. Table A.20 reports the variance decomposition of inflation from Section 3.1. Under fiscal dominance ($\psi_\pi < 1$), the ratio σ_R/σ_π is around 0.23. Under monetary dominance ($\psi_\pi > 1$), it drops to 0.01–0.05. Both ratios are significantly smaller than the 0.45 benchmark. These findings suggest that monetary policy shocks are moderate in our baseline, especially under monetary dominance.

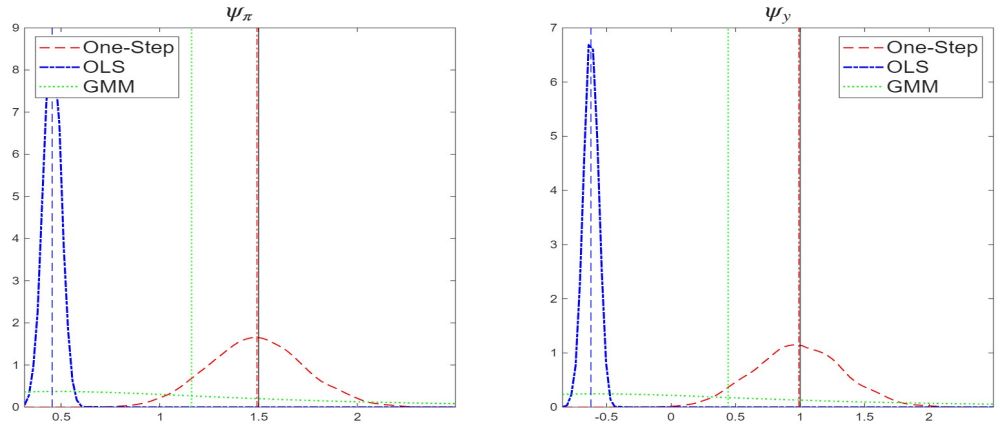
Table A.20: Variance decomposition of inflation

Fiscal dominance regime		Monetary dominance regime	
ψ_π	σ_R/σ_π	ψ_π	σ_R/σ_π
0.5	0.2336	1.05	0.0126
0.7	0.2342	1.2	0.0235
0.9	0.2347	1.5	0.0525

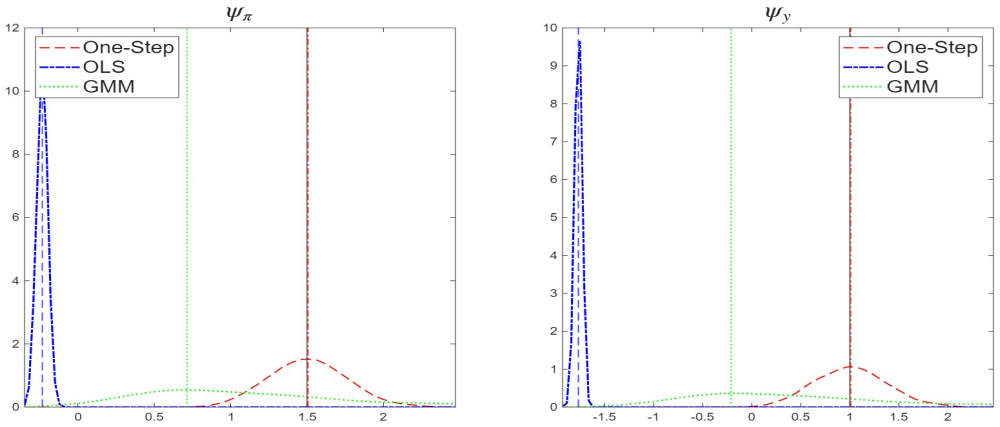
Notes: This table reports the ratio of monetary policy shock standard deviation to inflation standard deviation (σ_R/σ_π) under different regimes. The benchmark ratio of 0.45, calibrated by Carvalho et al. (2021) to match Smets and Wouters (2007), serves as our reference. Results indicate that monetary policy shocks contribute only modestly to inflation volatility in our baseline calibration. Other parameters are calibrated as in Table 1.

To further investigate the impact of monetary policy shock variance on the estimation results, we conduct simulations under three different levels of monetary policy shock variance: low ($\sigma_R = 0.2$), medium ($\sigma_R = 0.4$), and high ($\sigma_R = 0.7$). Figure 4 shows the estimation results under different levels of monetary policy shock variance.

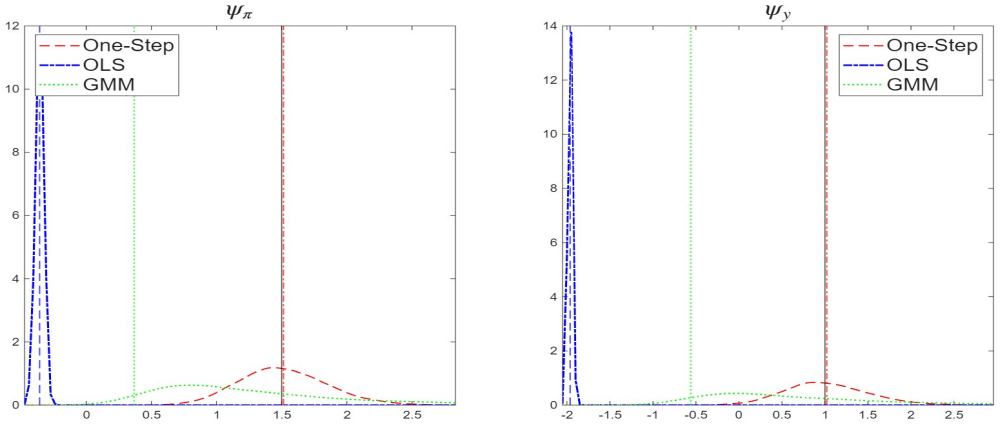
Results indicate the following. As the monetary policy shock variance rises, OLS estimates become biased. GMM estimates also become biased, but are less biased than OLS estimates, consistent with Carvalho et al. (2021). However, in our model, the variance of GMM estimates rises with monetary policy shock variance. Extreme outliers can lead to large biases. This shows that GMM estimation can be sensitive to fiscal-monetary interactions. In contrast, our one-step method remains unbiased at all variance levels. It reliably recovers the monetary policy parameters even with large shocks and fiscal-monetary interactions.



(a) $\sigma_R = 0.2$



(b) $\sigma_R = 0.4$



(c) $\sigma_R = 0.7$

Figure 4: Full information with different monetary policy shock variance

Notes: This figure illustrates the distribution of estimated monetarypolicy coefficients (ψ_π and ψ_y) obtained using different methods with different levels of monetary policy shock variance. The blue dashed-dotted line represents the distribution estimated by the OLS method, the green dotted line plots the distribution obtained from GMM, and the red dashed line corresponds to the distribution generated by the one-step method. The policy parameters are set as follows: $\psi_\pi = 1.5$, $\psi_y = 1$, and $\psi_s = 0$.